CRT anchors and smaller coefficients for ZFC consistency

Round 22 research report and reproducible artifacts

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Abstract

This round gives three complementary reductions of explicit polynomial presentations of syntactic ZFC consistency. Reusing an even CRT row-index representative as the semantic Pell anchor permits removal of a formula-label range test; weighted polynomial growth and exact integral projection then give a principal quartic in 215 unrestricted integers, with 7081 monomials and coefficient height 8880. A separate reciprocal-pooling composition reaches 212 unrestricted variables at degree 10. Monic product and chained-scale constructors reduce the smallest selected coefficient-height-one equation from 503 to 493 unrestricted integer variables. Finally, a full native-interface proof transports all three ten-positive-variable circuits to one coherent Start-only4127 coefficient tuple, reducing their four literal coefficient payloads from 295388 to 97046 bytes without changing the degree/cost pairs 214/87, 232/85 and 240/82.

Every equation is an ordinary fixed polynomial on its explicitly stated domain. The report provides complete witness arguments for the changes, prints every principal residual, and retains the native coefficient proof and all inherited source interfaces. The reproducible arrays and finite checkers establish exact identities and metrics; they do not replace the unbounded arithmetic or logical calibration. The result is a provable equivalence with consistency, not a proof of consistency, unsolvability, global minimality or a new end-to-end proof-assistant certificate.

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1 Results and precise comparison

The principal ordinary polynomial is

$$\boxed{P_{215}(x_0,\dots,x_{214}) = \sum_{j=0}^{153} R_j(x)^2 = 0, \qquad x \in \mathbb{Z}^{215}.} \tag{1}$$

Every residual is printed in Appendix A. Its finite literal array specifies every coefficient and independent coordinate; no variable exponent, oracle, recurrence, sequence or hidden witness is an algebraic primitive. The final collected polynomial has exact degree four, 7081 monomials and coefficient height 8880.

Theorem 1.1. *For the principal polynomial and each selected companion on its stated domain D, the ordinary witness arguments composed with the identified inherited syntactic calibration prove*

$$\mathrm{ZFC} \vdash [\neg\exists x \in D^n \; P(x) = 0] \iff \mathrm{Con}(\mathrm{ZFC}).$$

No consistency assumption is used to prove this equivalence.

This does not assert that either side is true. In particular it neither proves consistency nor decides whether an equation has a zero.

1.1 Direct quartics over unrestricted integers

Endpoint	Variables	Rows	Terms	Height
Round21 principal	217	154	7396	8880
Incoming CRT-anchor principal	215	154	5660	91679
Round22 principal	215	154	7081	8880
Round22 balanced	215	154	7315	4096
Round22 lower height	215	154	16077	2304
Round22 lower dimension	214	153	7078	1052028

Every row represents a distinct fixed quartic. The incoming anchor point is sparser than the selected moderate-height presentation; it is not claimed to be dominated. The 214-variable endpoint pays a substantially larger coefficient height, and the lower-height endpoint pays more support. The balanced equation uses 234 more monomials than the principal and reduces height from 8880 to 4096; the exact catalog preserves that tradeoff rather than hiding it in a single simplicity score.

The incoming source contributes the CRT-anchor and label-separation ideas. The new contributions are their actual composition with the transported quadratic simplifications, the weighted and quadratic growth budgets, and independently checked final coefficient arrays. Savings from different sources are not added without implementing their joint residual system.

1.2 A separate lower dimension tradeoff

The reciprocal companion has 212 unrestricted integer variables, 151 residuals, 55820 monomials and coefficient height 31104, but exact degree 10 rather than four. Its proof in Section 11 uses a cofinal $q \equiv 1 \pmod 3$ restriction, followed by reciprocal pooling of the q and $q+1$ certificates. The original even H is unchanged. Its higher degree and much larger height are explicit costs, not omitted coordinates in the quartic comparison. Against source 168's separate degree-ten point in 215 integers with 152 rows, 99914 monomials and height 113304, it improves all four displayed finite size measures; its arithmetic front is nevertheless different, not a relabeling of that source's zero set.

1.3 Height one and positive scalar equations

The selected height-one catalog ranges from 493/24828 to 701/7141 variables/monomials. Its nine points include the retained 557/11991 tradeoff. Each is a quartic with constant one and all collected nonzero coefficients in $\{-1, 1\}$, on unrestricted integer coordinates.

The scalar domain is instead exactly

$$(J, F, \alpha, i, T, w, s, n, \eta, \zeta) \in \mathbb{Z}_{>0}^{10}.$$

The three exact degree/live-operation pairs are 214/87, 232/85 and 240/82, unchanged from Round21. The new complete four-port tuple occupies 97046 minimal unsigned bytes instead of 295388, a 67.146 percent reduction. This is a raw numerical payload comparison, not compressed ZIP size, polynomial support or the height of the fully expanded scalar polynomial. Those latter two scalar measures are not claimed. The separate 11-unrestricted-integer six-port family is unchanged and is not a new selected artifact in this round.

2 Logical scope and literal polynomial definitions

2.1 What the consistency interpretation asserts

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}$ denotes the unrestricted integers. The domain is part of an equation, not a presentation detail. An equation over $\mathbb{N}$ is not thereby an equation over $\mathbb{Z}$, and an unrestricted construction requires its own sign argument or an explicit domain conversion.

Let $\mathrm{Ref}_{\mathrm{ZFC}}(r)$ be the fixed arithmetized predicate saying that r codes a finite syntactic ZFC refutation. For each selected endpoint P on the stated domain D, the claim is

$$\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} \; \mathrm{Ref}_{\mathrm{ZFC}}(r). \tag{2}$$

The ordinary transformations proved here are composed with the identified inherited calibration. On that proof chain, formalization in ZFC yields

$$\mathrm{ZFC} \vdash [\forall \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) \neq 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{3}$$

This report does not assert that either sentence in (3) is true. In particular it does not prove consistency, decide solvability, establish a global minimum, or supply a new end-to-end proof-assistant kernel check or encoded object-language ZFC derivation.

For the direct families, the retained ordinary logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The first relates the fixed numerical calculus to a finite ZF refutation; the second is the retained ZF-to-ZFC refutation equivalence. The arithmetic front implements that calculus using finite syntax constructors, reciprocal CRT carriers, power or factorial certificates as appropriate, and range tests. The selected scalar branch retains its separately identified native/compiler and refutation calibration at one coherent coefficient tuple. Its domain and calibration are not interchangeable with the unrestricted direct route.

2.2 The modular structure of the proof

The proof chain has four logically different layers.

1. The inherited calibration identifies a fixed finite arithmetic relation with syntactic refutation. Its provenance and exact numerical payloads are part of the source closure.

2. Unbounded arithmetic arguments prove both witness directions for the certificate relations. These include Pell classification, CRT choices, factorial fixed points, integer signs, and rank or interpolation lemmas.

3. Exact polynomial transformations replace one finite presentation by another. Integral substitutions, restored monic coordinates, graph identities and coefficient collection establish the claimed zero-set correspondence.

4. Finite independent checkers verify the actual emitted arrays and graphs, not merely an abstract instance of a theorem.

A hash certifies file identity, not a theorem; an exhaustive coefficient check certifies an identity, not an unbounded Pell assertion. Each layer is used only for what it establishes. The new sections give complete arguments for this round's transformations; the retained source closure supplies the inherited arithmetic and logical inputs at their stated boundary.

2.3 Explicit finite arrays define the direct equations

For a quartic endpoint, the ordered `variables` array in `equation.json` fixes the independent coordinates $x_0, \dots, x_{n-1}$. A term encoded as $[c, [i_1, \dots, i_d]]$ denotes $c \prod_{j=1}^{d} x_{i_j}$; the empty index list denotes the constant c. Repeated indices are retained and give ordinary powers. The m residual rows $R_0, \dots, R_{m-1}$ have integer coefficients and degree at most two. The equation is exactly

$$P(\boldsymbol{x}) = \sum_{j=0}^{m-1} R_j(\boldsymbol{x})^2 = 0. \tag{4}$$

Every coefficient and every coordinate appears in a finite supplied array. The adjacent `expanded_polynomial.json` stores the fully collected polynomial, including its constant term. There is no factorial, variable exponent, division, binomial coefficient, bounded quantifier, or sequence symbol among its algebraic primitives. Such expressions occur only in the proof explaining its zeros.

For either domain $D \subseteq \mathbb{Z}$, equation (4) is equivalent to $R_j(\boldsymbol{x}) = 0$ for every j, because each square is nonnegative. Exact degree four is established by a collected nonzero quartic coefficient, not merely by the upper bound $\deg R_j \le 2$. All independent coordinates are counted; named arithmetic abbreviations are not new unknowns.

The height-one endpoints are defined directly by their collected coefficient arrays. Their graph certificates prove the displayed energy identity and its zero-set correspondence. The independent checker reconstructs the energy and compares every collected coefficient. The height claim therefore concerns the final polynomial, not only its generating rows.

2.4 Arithmetic graphs define the scalar equations

The scalar graph leaves are the stated independent variables, ordinary integer constants, or fixed authentic numerical ports. The selected ten-positive-integer branch has its own strictly positive domain and complete coefficient tuple. Each internal node is binary addition, subtraction, or multiplication. An acyclic graph with one designated output therefore denotes a unique element of an ordinary polynomial ring over $\mathbb{Z}$. Registers abbreviate polynomials; they do not introduce hidden existential variables. There is no division or radical gate.

For the selected ten-positive branch the entire tuple β, K_0, M_C, M_F is the Start-only4127 tuple of Section 14, with $\beta = 2^{193969} - 1$. All four complete minimal unsigned numerical payloads are supplied and reconstructed from the finite tile language and coefficient recipe. The M_F port is already once shifted by β; it must not be shifted again. The earlier BFS943/radix67 positive tuple and the source34 six-port integer tuple are different objects and are not mixed with this tuple. No new eleven-integer endpoint is claimed in this report.

2.5 What each measure counts

For a nonzero collected polynomial $P = \sum_\alpha c_\alpha x^\alpha$, write

$$\deg P = \max_{c_\alpha \neq 0} |\alpha|, \qquad |\operatorname{supp} P| = \#\{\alpha : c_\alpha \neq 0\}, \qquad \operatorname{ht}(P) = \max_\alpha |c_\alpha|.$$

Support includes a nonzero constant coefficient. The number of residual rows is distinct from support and from the number of operations. Scalar cost counts live binary graph gates, including squares and multiplication by fixed constants; copying a register and reading a leaf are free. Dead nodes do not count. Unless an expanded scalar polynomial is materialized, its support and total coefficient height are unmeasured, not zero.

A point dominates another only when it is no worse in every displayed measure and strictly better in at least one, with domain and degree held fixed where required. This is a finite comparison among specified presentations. Height one, smaller dimension, sparse support and low graph cost are different objectives, and no single global simplicity ordering is claimed.

3 The retained proof-calculus architecture

The numerical calculus has truth code 0, falsity code 1, modus ponens, generalization, eight logical axiom schemata, and a guarded seed schema. Its literal square-shell constructors are

$$\begin{aligned} I(a,b) &= 2((a+b)^2+a), & M(a,b) &= I(a,b)+1,\\ A(a,b) &= 2((a+b)^2+3a+4b+3), & E(a,b) &= A(a,b)+1. \end{aligned}$$

They encode implication, membership, universal quantification, and equality. Put $N(a) = I(a,1)$ and $X(a,b) = N(A(a,N(b)))$. The pairing maps $Q_0 = (a+b)^2+a$ and $Q_1 = (a+b)^2+3a+4b+3$ enumerate complementary square shells; parity supplies the four constructors. The inherited decoding theorem transports finite refutations between these codes and the original Cantor-coded C_{222} calculus, fixing 0 and 1 [7].

For clarity the retained logical codes are

$$\begin{aligned} r_1 &= X(a,E(a,b)),\\ r_2 &= I(A(a,I(b,c)),I(A(a,b),A(a,c))),\\ r_3 &= I(E(a,b),I(M(c,a),M(c,b))),\\ r_4 &= I(A(a,A(b,c)),A(b,A(a,c))),\\ r_5 &= I(E(c,a),I(M(c,b),M(a,b))),\\ r_6 &= I(X(a,A(a,b)),b),\\ r_G &= I(E(c,a),I(E(c,b),A(M(a,b),E(a,b)))),\\ r_8 &= I(I(I(c,d),a),I(I(a,c),I(b,c))). \end{aligned}$$

Define

$$\begin{aligned} \mathrm{Req}(b,k,m) &= A(b,I(m,N(A(b,I(m,N(M(b,k))))))),\\ S(a,b,k) &= A(k,I(M(k,a),M(k,b))),\\ T(a,b,k,m) &= N(A(a,I(S(a,b,k),\mathrm{Req}(b,k,m)))),\\ \mathrm{Clos}(a,b,k,m) &= A(b,I(M(b,k),N(A(a,I(m,T(a,b,k,m)))))),\\ \mathrm{nks}(a,b,c) &= A(c,I(\mathrm{Clos}(a,b,c,c),N(\mathrm{Req}(b,c,c)))),\\ \mathrm{RV}(a,b,c) &= I(M(a,b),A(c,M(a,b))),\\ r_9 &= N(I(\mathrm{RV}(a,b,c),\mathrm{nks}(a,b,c))). \end{aligned}$$

The last schema is allowed when $c \neq a,b$. Its arithmetic guard is

$$G = (c-a)^2(c-b)^2-(d+1)^2 = 0, \qquad a,b,c,d \in \mathbb{N}.$$

Such a d exists exactly when the two disequalities hold. The parameter d can separately be an ordinary formula parameter in the r_8 branch. The alternatives do not require both interpretations simultaneously.

The proof transcript is encoded by positive weights $w(j) = 1 + Fj$. A backward axiom step has $S'w(r) = S$, modus ponens has $S'w(b) = Sw(a)w(I(a,b))$, and generalization has $S'w(A(a,b)) = Sw(b)$. Truth uses $S' = S$. Let $D_r = S'w(r) - S$ and define the analogous MP and generalization differences. The retained nonnegative local polynomial is

$$\mathcal{L} = (S' - S)^2 \prod_{r \in \mathcal{R}} D_r^2 (D_{r_9}^2 + G^2) D_{\mathrm{MP}}^2 D_{\mathrm{gen}}^2 + B_0^2 + B_1^2.$$

Here $\mathcal{R} = \{r_1, r_2, r_3, r_4, r_5, r_6, r_G, r_8\}$, and B_0, B_1 are the two beta-remainder equations. Thus $\mathcal{L} = 0$ means that one valid proof-rule branch and both beta equations hold. The principal construction retains this local polynomial unchanged.

4 Formula bounds without a range test

The old arithmetic source separately required a derived sum of formula labels to lie below the proof horizon in every local row. That required its own numerator power, binomial divisibility test, and Pell index. The replacement needs only a fixed, much more generous syntactic bound.

Lemma 4.1 (Constructor-depth bound)**.** *For $t \geq 16$, every formula label needed by a local rule with $0 \leq a, b, c, d < t$ is smaller than $t^{2^{20}-1}$. In particular it is at most $L = t^{2^{20}}$.*

Proof. For a positive integer s, if integer inputs $0 \leq x, y < s$, then $x, y \leq s - 1$ and each of $I(x,y), M(x,y), A(x,y), E(x,y)$ is less than $8s^2$. For the largest constructor, $A(s-1, s-1) = 8s^2 - 2s$, and E adds one. If the inputs are less than t^e, $e \geq 1$, each output is consequently less than t^{2e+1}. Starting with exponent one, a term of constructor depth h is bounded by $t^{2^{h+1}-1}$.

The exact syntax trees of the nine axiom codes have depths

$$4, 3, 3, 3, 3, 5, 4, 3, 19.$$

The MP and generalization labels have depth one; parameter labels have depth zero. The depth certificate checks the trees against the pinned literal constructor source, including all occurrences of N and X. The maximum depth nineteen proves the result. No asymptotic estimate or coefficient rounding enters this argument. □

Lemma 4.2 (Sharp product separation)**.** *Let L, t, F be positive integers with $L \geq t \geq 1$ and $F \geq L^t$. Consider two finite lists of positive integer labels at most L. Suppose the first has at most t entries and the second at most $2t + 1$ entries. If their products of weights $1 + Fj$ agree, their label multisets agree.*

Proof. First the lists have equal length. If their lengths differ, let the smaller length be $r \leq t$. The product for the smaller list is at most $L^r(F+1)^r$, while the larger is at least $(F+1)^{r+1}$. But $F + 1 > L^t \geq L^r$, a contradiction.

Write their common length as $m \leq t$. For $1 \leq k \leq m$, each elementary symmetric coefficient e_k of the labels satisfies

$$1 \leq e_k \leq \binom{m}{k} L^k \leq m^{m-k} L^k \leq L^m \leq F.$$

The middle estimate uses $\binom{m}{k} = \binom{m}{m-k} \leq m^{m-k}$. Thus the corresponding coefficients of the difference of $\prod(1 + jX)$ have absolute value strictly less than F; the constant coefficient is zero. If the polynomial difference were nonzero, divide its value at F by the lowest power with nonzero coefficient and reduce modulo F. A nonzero integer of absolute value less than F would be divisible by F, impossible. The two polynomials coincide. Their linear factors over $\mathbb{Q}$ recover every label with multiplicity. The empty-list case is immediate. Labels zero, if present, contribute unit factors and are discarded before this argument. □

The boundary $F = L^t$ is permitted. Bounding coefficient differences by twice the largest coefficient would miss this useful fact: both coefficients are positive, so their difference has absolute value at most $F - 1$.

The Round22 weighted family uses fixed positive integers c, h with $h \geq 16$, $4 \mid h$ and $ch^3 \geq 2^{20}$, and the semantic choices

$$t = h + 4z^2, \qquad T_2 = t^2, \qquad K = cT_2^2, \qquad F = t^K. \tag{5}$$

Then $K \geq 2^{20}t$, so $F \geq L^t$ exactly as required. The selected weighted source has $(c, h) = (32, 32)$. The dimension companion instead uses

$$t = 1024 + 4z^2, \qquad K = 1024t^2, \qquad F = t^K.$$

Again $K \geq 2^{20}t$, including equality at the smallest horizon. The separate T_2 coordinate is not needed in this companion. All horizons are cofinal multiples of four; all other arguments need only $t \geq 16$ and $F \geq t^t$. The actual K, not an obsolete exponent expression, is included in the square frame and orientation ledger. Only the polynomial horizon and exponent relations are direct rows; the power relation is certified by the retained Pell mechanism.

5 The factorial-free arithmetic front

Use the following semantic quantities:

$$B = \binom{F+t}{t}, \qquad N = B^x, \tag{6}$$

$$y = F(2v+1), \qquad C = 2c_0, \tag{7}$$

$$H = 2 + 4(C + a + b + c + d + S + S' + u + v_q), \qquad q = (BH)^x. \tag{8}$$

The nine large CRT representatives are $a, b, c, d, S, S', u, v_q, C$; their local residues encode one proof row. The independent names c_0 and v in (7) are unrestricted integer witnesses, not extra local schema parameters.

The endpoints and row-index congruence are

$$x = F + 1 + (y+1)u_0, \qquad x = 1 + [1 + (t+1)y]u_t, \tag{9}$$

$$B\eta = C + F + 1. \tag{10}$$

For a local row with index $i = C \bmod p^{ex}$, the beta equations are

$$x = S + [1 + (i+1)y]u, \qquad x = S' + [1 + (i+2)y]v_q.$$

The retained ranges are $a, b, c, d, C < t$, $S, S' \leq y$, and $u, v_q < x$ on local decoded residues. There is no longer a range test on a derived formula-label sum.

5.1 Why one binomial modulus covers every row

Here is the inherited selector theorem in the form used by this construction [2]. Let $\Lambda_t = \operatorname{lcm}(1, \ldots, t)$. If $R \geq \Lambda_t^2$ and $B = \binom{R+t}{t}$, then:

1. for every $1 \leq i \leq t$, some full prime power $p^e \| B$ divides $R + i$ and satisfies $p^e > t$;
2. every full prime power $p^e \| B$ divides $R + i$ for at least one such row i.

Both directions matter. The first authenticates each row; the second makes the reverse CRT construction compatible at every prime of B.

For completeness, the elementary proof is short. The valuation formula shows $t\binom{t-1}{i-1} \mid \Lambda_t$. Since $R+i > \Lambda_t^2$, some prime has $w = v_p(R+i) > 2a$, where $a = \lfloor \log_p t \rfloor$. Other factors $R+j$ have the same p-valuation as $j-i$. Consequently

$$v_p(B) = w - v_p\left(t\binom{t-1}{i-1}\right) > a, \qquad v_p(B) \le w,$$

which proves the first statement. For the second, let $w = \max_i v_p(R+i)$. In Legendre's formula for $v_p(B)$ every summand is zero or one, and all levels above w vanish. Hence $v_p(B) \le w$; a row attaining the maximum is compatible.

The standard elementary estimate $\Lambda_t \le 4^{t-1}$ gives $F \ge t^t \ge 16^t > \Lambda_t^2$. Therefore $R = F$ is valid. Unlike the parent, this construction has no equation requiring a variable to equal a factorial.

5.2 The range lemma and precision

For a prime p, integers $A \ge 0$ and $k, e \ge 1$,

$$p^e \mid \binom{A}{k} \quad \Longrightarrow \quad A \bmod p^e < k. \tag{11}$$

If none of the k consecutive factors $A-k+1, \dots, A$ is divisible by p^e, the valuation formula has at most $e-1$ nonzero levels, a contradiction. A factor divisible by p^e gives the claimed residue. The convention $\binom{A}{k} = 0$ for $A < k$ covers the boundary.

At a selector $p^e \| B$, the range for C and (10) give the exact local index $i-1$ in $0, \dots, t-1$. The other ranges recover the bounded local tuple modulo p^{ex}. The retained local bound is

$$0 \le \mathcal{L} < (2+t+x+y+F)^{2^{23}} < 2^x \le p^{ex}. \tag{12}$$

Its first inequality is the existing literal C_{222} circuit majorant [6]; changing the size of F does not change that polynomial bound. Once signs are recovered, $x \ge 2+(t+1)y$ and $y \ge F$ imply $2+t+x+y+F < 2x$ and $x \ge F+1 > 2^{32}$. If $2^n \le x < 2^{n+1}$, $n \ge 32$, then $2^{23}(n+2) < 2^n \le x$, proving the second strict inequality in (12). Thus $N \mid \mathcal{L}$ forces each selected local value to be zero.

6 A freely chosen CRT index as the Pell anchor

The incoming anchor change is independent of the detailed syntax of C_{222}. An integer used as a CRT representative can be enlarged by whole multiples of its modulus without changing the finite transcript it represents. We use that freedom to make C the largest requested Pell index.

Write $X_j(A), Y_j(A)$ for the Pell polynomials, defined by

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{j+1} = AX_j + (A^2-1)Y_j, \quad Y_{j+1} = X_j + AY_j.$$

Signed indices have $X_{-j} = X_j$, $Y_{-j} = -Y_j$. The square frame is $W = 2 + \sum \alpha^2$, where every actual power base, index, and proposed output occurs among the α. The retained odd Pell frame proves $|A| > W^W$ and $|A-1| > 4W^3$.

The fourteen nominal indices are

$$t,\ F+t,\ a,b,c,d,\ S,\ y+1,\ S',\ u,\ x,\ v_q,\ C,\ K. \tag{13}$$

The former derived-label pair is reassigned to K; the existing C pair becomes the anchor. With anchor coordinates (U, Y), retain auxiliary Pell pairs (J_X, J_Y) and (V_X, V_Y) and comparison center D satisfying

$$U^2 - (A^2-1)Y^2 = 1, \qquad\qquad J_X^2 - (A^2-1)J_Y^2 = 1,$$

$$V_X^2 - (D^2-1)V_Y^2 = 1, \qquad J_Y = Y^2(2e+1),$$
$$D = 1 + 4J_Y(2d_0+1), \qquad D = A + J_X f_0,$$
$$V_X = U + J_X g_0, \qquad V_Y = C + 4Y h_0.$$

The squared bound is

$$Y^2 = C^2 + 1 + \sum_{i \neq \text{anchor}} Y_i^2 + r^2 + s^2 + w(w+1). \tag{14}$$

All these variables are unrestricted integers. Multiplication wires turn the displays into quadratic residuals; the final emitted data contain the actual wires, not a higher-degree shorthand.

Proposition 6.1 (Anchor substitution)**.** *The retained signed anchor and pooled orientation proof applies with nominal index C in place of the derived formula-label index. No monotonicity or syntactic interpretation of C is required in soundness. For completeness, C can be chosen even and larger than all other indices in* (13) *while preserving every required CRT residue.*

Proof. Equation (14) gives $|Y| > |C|$, so $Y \neq 0$. Pell classification and the retained squared-divisibility argument give an absolute index $i > 0$ with $|Y| = Y_i(|A|)$ and $i = \pm C \pmod{2|Y|}$. Since $1 \leq i \leq |Y|$ and $|C| < |Y|$, one obtains $C \neq 0$ and $i = |C|$. Every other classified index is strictly smaller than $|C|$ by the same squared bound.

The pooled orientation row is, modulo its retained norm identities,

$$\sum_i \big((X_i - 1)^2 + (Y_i - k_i)^2\big) = (A-1)Q_0.$$

Modulo $A-1$, write $s_i \in \{-1, 1\}$ and r_i for the small abscissa and ordinate residues of each classified pair. The absolute value of r_i is its classified index, hence at most $|C| < W$. Thus $|s_i - 1| \leq 2$ and $|r_i - k_i| < 2W$. For the fourteen pairs their squared-error sum satisfies

$$0 \leq \sum_i \big((s_i - 1)^2 + (r_i - k_i)^2\big) < 4 \cdot 14(W^2+1) < 4W^3 < |A-1|.$$

It must vanish, identifying every pair with the canonical signed pair at its nominal index. This includes the new K pair and the C anchor. These are exactly the hypotheses and steps of the inherited signed-anchor proof [5]; the name or origin of the nominal index never enters.

For the reverse construction, first fix t, K, F, y, x, B, N and the prescribed residue classes of the nine representatives. None of these choices depends on the eventual representative C. Since t is divisible by four and $F = t^K$, $v_2(F) = Kv_2(t) > \lfloor \log_2 t \rfloor$. Therefore $v_2(F+i) = v_2(i)$ for $1 \leq i \leq t$, and B and N^2 are odd. Each prescribed class modulo N^2 has cofinally many positive even representatives, with step $2N^2$. Choose the other eight representatives first, then choose C in its class above all the other fixed indices. This also preserves $B \mid C + F + 1$, since $B \mid N^2$. Only afterward are H, q, W, A chosen. There is consequently no cyclic size requirement.

For canonical pairs at a positive sufficiently large A, the strict index inequality makes the slack in (14) nonnegative. Since C is even, $Y_C(A)$ is even and $X_C(A)$ odd. The inherited explicit auxiliary choice at index $CY_C(A)$ makes $J_Y/Y_C(A)^2$ odd; a two-modulus CRT then constructs D and the remaining comparison coordinates. Every nonnegative slack has the ternary representation $r^2 + s^2 + w(w+1)$. Thus all anchor witnesses exist. □

7 Power certificates and the noncircular sign argument

The polynomial retains sixteen power requests: three individual requests for $F = t^K$, $N = B^x$, $q = (BH)^x$; ten at base $q+1$ for the binomial numerators; and three at base q for q^t, q^{y+1}, q^x. The incoming square frame has thirty-five actual atoms, including duplicates where the same

integer has two roles. Its coefficient identity is independently checked against every requested base, index, and output.

For a block at nonzero base n, let $Z_i = AY_i$ and let m_i be the proposed output. The inherited pooling mechanism uses just two rows:

$$K_n - 2\sum_i Y_i(m_i - X_i) - (2A - n)L_n = 0,$$
$$\sum_i ((m_i - X_i + Z_i)^2 - Y_i^2) - nK_n + L_n = 0.$$

They imply divisibility of the sum of squared power errors by $T_n = 2An - n^2 - 1$. To handle signed indices, put $b = \lfloor \sqrt{W - 1} \rfloor$ and multiply by n^{2b}. Canonical Pell congruences then give

$$T_n \mid E = \sum_i (m_i n^b - n^{b+k_i})^2, \qquad 0 \le E \le 4rb^{4b} < W^W < |T_n|.$$

Here $r \le 15$, every exponent $b + k_i$ is nonnegative, and $\gcd(n, T_n) = 1$. Consequently every output is exact; at $|n| \ge 2$ the index is nonnegative, since a negative index would force an integer times a positive power of $|n|$ to equal one. The individual requests use the same canonical power argument. All displayed exponentials occur in this proof, not as operations in the final polynomial.

It remains essential to justify the bases before interpreting these requests. The order is as follows.

1. The known $t \ge 16$ and positive K give $F = t^K > 0$, even. Since $C = 2c_0$, equation (10) has odd right side, so $B \ne 0$. Independently, $H \equiv 2 \pmod 4$ for every signed representative tuple. Thus $|BH| \ge 2$.

2. The individual BH request gives $x \ge 0$ and $q = (BH)^x$. Equation $y = F(2v + 1)$ gives $|y| \ge F$. If $x = 0$, the last endpoint would make the integer $1 + (t + 1)y$, whose absolute value exceeds one, divide -1. Therefore $x \ge 1$, $|q| \ge 2$.

3. The q block is now legitimate. Its $y + 1$ index is nonnegative. A negative y would be at most $-F < -1$, so $y \ge F$. The first endpoint excludes $u_0 < 0$, and the last excludes $u_t \le 0$. Thus $x \ge F + 1$ and $x \ge 2 + (t + 1)y$.

4. Hence $|q + 1| \ge 2$. Its block recovers the nonnegative signs of all nine representatives and all their exact numerator powers. Now $H \ge 2$. No earlier step needed these signs.

5. The first binomial digit authenticates $B > 0$, as proved next. It follows that $q > 0$ and $N = B^x > 0$, completing the sign recovery.

7.1 Exact recovery of the digits

Let $M = q^x$. Ten normalized remainder rows share

$$M^2 = 1 + 4\sum_{i=1}^{10} \rho_i^2 + \alpha^2 + \beta^2 + \gamma(\gamma + 1). \tag{15}$$

For a denominator $D_i = q^{k_i}$, the ratio equation $D_i a_i = M$ and normalized digit equation imply $r_i = \rho_i / a_i \in \mathbb{Z}$ and the usual unscaled division equation. The norm gives $|r_i| < |D_i|/2$, even before signs of the radix are known. The selected Round22 source retains this coefficient-four sphere exactly; an exploratory wider sphere is not part of the release. A true low binomial remainder with absolute value below $|D_i|/32$ must equal r_i, by congruence and the strict interval width.

For the first digit, every coefficient of $(q + 1)^{F+t}$ through degree t is at most $(F + t)^t \le 2^{t(F-1)}$. On the other hand, $|q| \ge 2^x \ge 2^{(t+1)F+2}$. The true low remainder and the t-th coefficient are therefore smaller than their needed denominators divided by 32. The normalized first row gives

$B \equiv \binom{F+t}{t} \pmod q$. Also $|B| \le |BH| < |q|/2$, since $x \ge 3$; the positive binomial coefficient is below $|q|/32$. This proves the exact identity $B = \binom{F+t}{t} > 0$.

For every other numerator with upper argument $a \ge 0$ and requested digit level $k \le x$, $\binom{a}{j} \le \max(1,a)^x \le H^x$. Since $q = B^x H^x$ and $B \ge 2$, the same strict remainder bound follows. There is no derived-label numerator needing a separate degree estimate here. The ratios prove the necessary denominator divisibilities, and the decoded digit equations yield the nine retained range predicates.

8 Soundness and completeness of the new front

8.1 From an integer zero to a finite refutation

The previous sections recover every sign, power, binomial digit, and local range from an arbitrary integer zero. The selector theorem and (12) supply exactly t valid local proof steps. Their beta states join because the same x, y determine the unique remainders modulo $1 + (i+1)y$.

The possible boundary $y = F$ must not be overlooked. Then the first beta remainder is zero, whereas the last is one. Every valid local weight equation propagates a zero state to zero, since all weights are positive. This contradiction proves $y > F$. Thus the initial and terminal states are respectively $F + 1 = w(1)$ and 1.

Multiplying the local equations and cancelling the nonzero intermediate states gives equality between the multiset of conclusion weights and the initial falsity weight together with all premise weights. There are at most t nonunit conclusion factors and at most $2t+1$ nonunit premise factors. All local labels are at most L by Lemma 4.1. Lemma 4.2 gives equality of the corresponding label multisets.

Finally, a balanced single-conclusion rule flow implies a finite derivation of its distinguished target. To see this without assuming the flow is acyclic, close the finitely many displayed rules under derivability, starting with the valid axiom and truth conclusions. If falsity were outside that closure, every rule with an outside conclusion would have at least one outside premise. Summing "outside conclusions minus outside premises" over all occurrences would be at most zero. The recovered balance makes the sum exactly one, contributed by the target falsity. Contradiction. The finite closure computation itself extracts a derivation.

8.2 From a finite refutation to an integer zero

Start with a finite backward proof-tree expansion. Choose the cofinal horizon $t = h + 4z^2$ (or $t = 1024 + 4z^2$) larger than its length and every local schema parameter, including all freshness witnesses. There is no need to make t exceed the full expanded formula codes. Set $K = ct^4$ (or $K = 1024t^2$) and $F = t^K$; form all finite pending-multiset products with weights $1 + Fj$, and pad the transcript with truth steps to length t.

Let O be the odd part of $t!$. Choose an arbitrarily large positive odd multiple w of O and put $y = Fw$. Since $v_2(F) = Kv_2(t) \ge 2t > v_2(t!)$, one has $t! \mid y$. Choose w large enough that $y > F$ and every pending state is at most y. The moduli $1 + (i+1)y$, $0 \le i \le t$, are pairwise coprime: a common prime would divide both y and one of these moduli after taking an index difference at most t. CRT gives x with the prescribed state remainders; its residue class is cofinal, so all needed endpoint and quotient inequalities can be met.

Now set $B = \binom{F+t}{t}$ and $N = B^x$. At every full prime power $p^e \| B$, select one compatible row and choose all nine representatives to match that row modulo p^{2ex}. The generalized CRT theorem justifies this construction at every prime, including primes not used as the distinguished selector of any row. Polynomial evaluation preserves the local congruences. For a range with local value $w < k$, the falling factorial contains the factor $A - w$, divisible by p^{2ex}; the denominator loses at most $v_p(k!) \le k - 1 < x \le ex$ powers. Thus the required binomial coefficient is divisible by p^{ex}.

Choose the eight ordinary representatives first. Then choose C even, positive, and larger than every other nominal index, in its unchanged class modulo the odd integer N^2. Define H and q from their actual values, so enlargement of C strengthens the radix bounds. The ten true normalized remainders are each less than $M/32$. Their common slack in (15) is nonnegative, because $4 \cdot 10/32^2 < 1$. Every integer $n \geq 0$ has the form $r^2+s^2+w(w+1)$: apply the three-square theorem to $4n+1$, whose representation has one odd and two even coordinates.

Choose the large positive Pell center from the square frame, all canonical pairs, and the even-C auxiliary witnesses of Proposition 6.1. The canonical congruences fill the individual and pooled power quotients. This constructs an integer zero of the complete retained quadratic source with the chosen growth rows. The exact projections below carry it to the selected final equation.

9 Exact power pooling and integral algebra

9.1 The inherited two row certificate

For $|A| > 1$ let $(\chi_A(k), \psi_A(k))$, $k \in \mathbb{Z}$, be the canonical signed Pell pair. Its recurrence matrix is $\left(\begin{smallmatrix} A & A^2-1 \\ 1 & A \end{smallmatrix}\right)$, with initial pair $(1, 0)$. Its determinant is one, so all signed indices are defined. Given a nonzero base n, put

$$T_n = 2An - n^2 - 1.$$

Then $\gcd(n, T_n) = 1$, and

$$\chi_A(k) + (n - A)\psi_A(k) \equiv n^k \pmod{T_n} \qquad (k \in \mathbb{Z}). \tag{16}$$

Indeed $(n - A)^2 \equiv A^2 - 1 \pmod{T_n}$; evaluating the fundamental unit at $n - A$ gives n. Positive powers follow by multiplication, and negative powers by invertibility. The Pell functions are proof notation, not primitives in the final polynomial.

For r requests (k_i, m_i) at this base, retain the existing coordinates $X_i = \chi_A(k_i)$, $Y_i = \psi_A(k_i)$ and $Z_i = AY_i$. Introduce only two new integer coordinates K, L, and impose

$$K - 2\sum_i Y_i(m_i - X_i) - (2A - n)L = 0, \tag{17}$$

$$\sum_i ((m_i - X_i + Z_i)^2 - Y_i^2) - nK + L = 0. \tag{18}$$

Every term has degree at most two. Neither T_n nor any displayed summation abbreviation introduces another independent coordinate.

Lemma 9.1. *The two rows imply*

$$\sum_i \big(m_i - X_i + (A - n)Y_i\big)^2 = T_n \left(L - \sum_i Y_i^2\right).$$

Conversely, when each error on the left is divisible by nonzero T_n, there are integers K, L satisfying both rows.

Proof. Set $D_i = m_i - X_i + AY_i$ and $H = \sum_i Y_i(m_i - X_i)$. Eliminating K gives $\sum_i (D_i^2 - Y_i^2) - 2nH = T_n L$. Expanding $\sum_i (D_i - nY_i)^2$ subtracts exactly $T_n \sum_i Y_i^2$, proving the identity. For the converse choose

$$L = \sum_i Y_i^2 + T_n^{-1} \sum_i (D_i - nY_i)^2, \qquad K = 2H + (2A - n)L.$$

The divisibility assumption makes L integral and direct substitution proves both rows. No sign of K or L is required. □

Theorem 9.2 (Simultaneous exactness from a square frame)**.** *Suppose $W \geq 258$, $1 \leq r \leq 15$, $n \neq 0$, $|A| > W^W$, and $n^2, k_i^2, m_i^2 < W$. If the pairs are the specified canonical signed Pell pairs, the two quadratic rows force $m_i = n^{k_i}$ for every i. For $|n| \geq 2$ they also force $k_i \geq 0$; for $n = \pm 1$ all signed indices are allowed. Conversely all these exact powers admit K, L.*

Proof. Put $b = \lfloor \sqrt{W-1} \rfloor \geq 16$. Every absolute base, index and proposed output is at most b. Multiply the aggregate congruence by n^{2b}, using (16), to get

$$T_n \mid E, \qquad E = \sum_i (m_i n^b - n^{b+k_i})^2 \geq 0.$$

All exponents are nonnegative. Since $b \geq 2$, $|m_i n^b - n^{b+k_i}| \leq b^{b+1} + b^{2b} \leq 2b^{2b}$. Consequently

$$E \leq 4r b^{4b} < W^{2b+1} \leq W^W,$$

because $4r \leq 60 < W$, $b^2 < W$ and $2b + 1 \leq b^2 + 1 \leq W$. On the other hand $|T_n| \geq 2|A| - W > |A| > W^W$. Divisibility forces $E = 0$, so every summand vanishes. A nonnegative k_i gives the asserted power by cancelling n^b. If $k_i = -j < 0$, cancellation gives $m_i n^j = 1$; this excludes nonunit bases and gives the stated power at the two units. For the converse each exact error is divisible by T_n, so apply Lemma 9.1. □

The absolute-value argument includes negative radices and the unit case; assuming a positive radix before decoding its sign would not suffice. The actual square-frame ledger, canonical-pair classification and request ports are bound to the literal source by the finite checker. A different affine frame that only gives $|a| < W$ cannot silently be substituted for this square-frame hypothesis.

9.2 Exact integral projections along quadratic radicals

Write each residual as $Q_i = q_i + \ell_i + c_i$, with homogeneous quadratic part q_i, linear part ℓ_i, and integer Hessian H_i. For an integer vector v with $H_i v = 0$ for every i, exact expansion gives

$$Q_i(y + sv) = Q_i(y) + s\ell_i(v). \tag{19}$$

The mixed term is $sy^T H_i v = 0$ and the pure term vanishes since $2q_i(v) = v^T H_i v = 0$.

Theorem 9.3. *Suppose $v_p = 1$ and one slope $a_r = \ell_r(v)$ is ± 1. With $y_p = 0$, let*

$$R_i(y) = Q_i(y) - a_i a_r Q_r(y), \quad i \neq r, \qquad a_i = \ell_i(v).$$

These integer quadratics have a common zero set in bijection with that of all Q_i, with one fewer variable and residual. The lift is $x = y - a_r Q_r(y) v$, and the inverse is $y = x - x_p v$.

Proof. Put $s = -a_r Q_r(y)$. Equation (19) gives $Q_r(y + sv) = 0$ and $Q_i(y + sv) = R_i(y)$ for $i \neq r$. Conversely an old zero gives $y_p = 0$ and $Q_r(y) = -x_p a_r$, so the displayed lift restores precisely x_p. All other $R_i(y)$ vanish by the same formula. The two maps are inverse and integral; the new rows are constant combinations of restricted quadratics. No division condition remains. □

A primitive direction alone would not justify division by a nonunit slope. The actual certificates require the two literal unit conditions. These unrestricted-integer coordinate changes need not preserve a positive domain.

Lemma 9.4 (Affine consequence elimination)**.** *Suppose a nonzero constant integer combination $L = \sum_i \lambda_i Q_i$ has degree at most one and its primitive affine normalization has coefficient ± 1 on x_p. Solve it as $x_p = S(\widehat{x_p})$. Substitute in every row and omit any row r with $\lambda_r \neq 0$. The resulting quadratic system has an integer zero set in bijection with the original system.*

Proof. Every original zero satisfies $L = 0$, hence $x_p = S$ and every retained substituted row. Conversely, the substitution makes L identically zero. At a zero of the retained rows, $\lambda_r Q_r = 0$; the integers have no zero divisors, so $Q_r = 0$. Insertion of S and deletion of x_p are inverse. A unit λ_r is not needed; a unit solved variable coefficient after primitive normalization is essential. □

An elementary integer row operation $Q_i \leftarrow Q_i + kQ_j$, $i \neq j$, has inverse with $-k$. A triangular integral affine coordinate change has its explicit integral inverse. A monic quadratic definition can be eliminated only when its complete substitution retains the claimed degree. Each released certificate records these conditions and the exact index compaction. A sum of squares of the final rows is generally a *different* polynomial, with the same zero-existence statement.

10 Weighted growth and the emitted integer charts

The weighted source has $(h, c) = (32, 32)$ in (5); the quadratic companion has $t = 1024 + 4z^2$ and $K = 1024t^2$. Both attain the admissible boundary $K/t \geq 2^{20}$. Unlike a mere change of names, the actual defining residuals, frame atom and requested Pell index are all changed together and checked against the retained source.

10.1 A literal local majorant without a derived range

The removal of the derived-label range also requires an independent bound for the actual local Horn polynomial. Here is an explicit circuit argument. Put $R_0 = 2 + t + x + y + F$. On a decoded local tuple, the genuine inputs $t, x, F, y + 1, a + 1, b, c, d, S, S', u, v_q, C$ are smaller than R_0; here C is the *local residue*, not its enlarged global representative. The extra product FS is bounded by R_0^2.

For a monic gate $v = \sum_m c_m \prod_j v_{mj}$, suppose $|v_{mj}| \leq R_0^{e_{mj}}$ and there are n nonzero summands. Since $R_0 \geq 2$, the triangle inequality gives

$$|v| \leq R_0^e, \qquad e = \max_m \left(\lceil \log_2 |c_m| \rceil + \sum_j e_{mj} \right) + \lceil \log_2 n \rceil.$$

Induction follows the actual monic dependencies, not a guessed syntactic name. The independent ledger checks coordinates 20 through 99 using rows 5 through 83 and 85 of the literal parent225 circuit. Removing exactly the N-quotient monomial from row 84 isolates the local constraint. Its resulting exponent is $5339613 < 2^{23} = 8388608$. Thus the local nonnegative constraint is below $R_0^{2^{23}}$ without the removed range test. As already proved, $R_0 < 2x$ and $x > 2^{32}$ imply $R_0^{2^{23}} < 2^x$. This is precisely the strict modulus margin needed for exact decoding, and is independent of the inflated global C.

10.2 More detail on the signed anchor witnesses

For completeness we spell out the two sensitive divisions in the anchor argument. In the notation of Proposition 6.1, principal Pell classification at $|A|$ gives positive indices i, l with $|Y| = Y_i(|A|)$ and $|J_Y| = Y_l(|A|)$. Strong divisibility gives $i \mid l$. Write $l = is$. Composition gives

$$Y_l(|A|) = Y_i(|A|) Y_s(X_i(|A|)), \qquad Y_s(X_i) \equiv s X_i^{s-1} \pmod{Y_i}.$$

The Pell norm makes X_i a unit modulo Y_i. Since $Y_i^2 \mid Y_l$, we get $Y_i \mid s$ and consequently $|Y| \mid l$. The independent abscissa comparison gives $j \equiv \pm i \pmod{2l}$ for the comparison index, while $D \equiv 1 \pmod{4Y}$ and $V_Y \equiv C \pmod{4Y}$ give $j \equiv \pm C \pmod{4|Y|}$. Hence $i \equiv \pm C \pmod{2|Y|}$. Together with $1 \leq i \leq |Y|$ and $|C| < |Y|$, this proves $C \neq 0$ and $i = |C|$. Neither nonnegativity nor a formula-label interpretation of C is used.

Conversely choose the positive canonical center and even C. With $Y = Y_C(A)$, $U = X_C(A)$, use $l = CY$ and $J_Y = Y_l(A)$. Composition and the polynomial expansion modulo Y^2 show

$$J_Y/Y^2 \equiv U^{Y-1} \pmod{Y}.$$

Since Y is even and U odd, this quotient is odd. Thus $e = (J_Y/Y^2 - 1)/2$ is integral. The index l is even, so J_X is odd; the norm gives $\gcd(J_X, 8J_Y) = 1$. CRT therefore solves

$$D \equiv A \pmod{J_X}, \qquad D \equiv 1 + 4J_Y \pmod{8J_Y}$$

with $D \geq 2$. Evaluating the Pell polynomials at D gives all remaining comparison quotients integrally. The dominant index makes the cap slack nonnegative, and its ternary-square representation completes the cap. These are actual integer witnesses, not a divisibility assertion with an unjustified quotient.

10.3 Exact transport followed by two radical projections

The Round21 chart supplies proposed operations, not a semantic theorem about the new coordinate names. The literal emitted sequence retains only operations checked on the actual source:

1. Integer shears $x_j = y_j + ky_i$, $i \neq j$, with inverse $y_j = x_j - kx_i$.
2. Integer row additions $R_i \leftarrow R_i + kR_j$, with inverse obtained by replacing k by $-k$.
3. Monic definitions $x_v = f(\widehat{x_v})$, followed by deletion of that row and coordinate. A quadratic restoration is used only when the old coordinate occurs linearly everywhere else and every final row remains quadratic. Inserting f is the exact integral inverse on zeros.

Each profile then admits two common-Hessian projections from Theorem 9.3. Their actual direction and row certificates satisfy the literal unit conditions, and the independent checker reconstructs both directions without importing the producer. Weighted growth gives 215 coordinates and 154 rows; quadratic growth gives 214 coordinates and 153 rows. Subsequent support/height changes are invertible integer row operations. The sum of their squares may change away from the common zero set; no identity of those sums is asserted.

The source change for the quadratic-growth companion deletes precisely the private $T_2 = t^2$ coordinate and defining row, while changing $K = cT_2^2$ to $K = 1024t^2$. The existing request is at actual K and the frame keeps that actual value. This is not deleting a wire while leaving its old occurrences in the circuit.

10.4 The inherited calibration and literal identity pins

The parent225 residual array has SHA-256

`fee84c408828aba3e7ceebe9a37f58509327063aa7fd078fd49c80adf3af4e56`.

The inherited anchor221 array has SHA-256

`4024d913bde43aa47e8be1fe8c4f36f560420e8b5ceef999d1625e2e31c31c3e`.

The complete source dependency boundary and exact inventory accompany these arrays. The ordinary calibration uses `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`, pinned in the incoming source to commit `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`. The new proof transfers these statements by unbounded witness arguments and exact algebra, rather than claiming that a finite coefficient test establishes consistency or a new complete formalization.

11 A separate degree ten companion in 212 integers

The reciprocal companion is not a quartic. It has exactly 212 unrestricted integer coordinates, 151 residuals, exact total degree 10, 55820 collected monomials and coefficient height 31104. Its largest residual has degree five and coefficient height 108. The polynomial is the sum of the squares of its complete literal residual array; its variable order and coefficients are supplied separately from the unchanged principal quartic.

The incoming reciprocal-pooling theorem comes from source 168. Its old factorial-based sign bootstrap is not imported into the present factorial-free front. Instead we prove a cofinal restricted chart in which both bases are nonzero before the power requests are interpreted; unit bases are allowed by the reciprocal theorem and are eliminated only after source restoration. The original $H = 2 + 4(C + \sum A_i)$ remains unchanged.

11.1 A cofinal radix chart with no early sign assumption

Restrict the source by the literal substitution

$$q = 3z + 1, \qquad z \in \mathbb{Z}.$$

At every integer input either $q \geq 1$, $q + 1 \geq 2$, or $q \leq -2$, $q + 1 \leq -1$. Both bases are therefore nonzero and have the same sign, before any semantic power interpretation. The possible values $q = 1$ and $q + 1 = -1$ are not prematurely excluded. This is an integral chart of a restricted subfamily, not a unimodular change on all old radix values. We must prove that the restriction preserves completeness.

Choose the horizon $t = 32 + 4a^2$ with $a \not\equiv 0 \pmod 3$, for example $a = 3v + 1$ arbitrarily large. Then $12 \mid t$, and these horizons remain cofinal. Put $K = 32t^4$ and $F = t^K$, in that order; the power is not K^t. For $p = 2, 3$ set $v_p(F) = v$ and $v_p(t!) = a_p$. Then $v > \max_{1 \leq i \leq t} v_p(i)$. Expanding the numerator of the binomial gives

$$t!(B - 1) = \prod_{i=1}^{t}(F + i) - t! = \sum_{\varnothing \neq S \subseteq \{1,\ldots,t\}} F^{|S|} \prod_{i \notin S} i.$$

Each summand has valuation

$$a_p + \sum_{i \in S}(v - v_p(i)) \geq a_p + 1.$$

Canceling the known p-power and the remaining unit part of $t!$ gives $B \equiv 1 \pmod p$. This is an integer valuation proof, not reduction of uncanceled rational factors modulo p. In particular $\gcd(B, 6) = 1$.

As in the main completeness proof, choose the finite path, then sufficiently large $y = Fw$ with odd w and $t! \mid y$. Every beta modulus $1 + (i + 1)y$ is odd. Their prescribed CRT class therefore has arbitrarily large even representatives; choose an even x satisfying all retained bounds. Now $N = B^x$ is coprime to six. Fix the eight ordinary representatives in their prescribed classes modulo N^2. Because $\gcd(N^2, 6) = 1$, the remaining class for C can be chosen even and in either of the two residues modulo three for which $3 \nmid H$. Its valid values form a cofinal progression of step $6N^2$. Choose C above all other nominal indices and representatives. This preserves all previous CRT congruences and $B \mid C + F + 1$.

Only now define H and $q = (BH)^x$. Here $H \equiv 2 \pmod 4$, $3 \nmid H$, $B \equiv 1 \pmod 3$, and x is positive even. Consequently $q \equiv 0 \pmod 4$ and $q \equiv 1 \pmod 3$, hence $q \equiv 4 \pmod{12}$. In particular $q \equiv 1 \pmod 3$, so the coordinate $z = (q - 1)/3$ is integral. The stronger modulo-twelve conclusion is useful only for constructing canonical witnesses, not imposed on arbitrary source zeros. Finally choose the actual square frame, Pell center and even-anchor cap witnesses as before. The choice order has no cyclic dependence:

$$t, K, F, \text{path}, y, x, B, N, \text{other representatives}, C, H, q, \text{frame}, A.$$

Evenness of x is a canonical completeness choice, not an assumption on arbitrary soundness witnesses.

11.2 The reciprocal polynomial and its exact power theorem

For $n_0 = q, n_1 = q+1$ let I_j denote the two request groups, containing three and ten requests. Retain the actual canonical Pell pairs (X_i, Y_i), outputs m_i, indices k_i, and wires $Z_i = AY_i$. Set

$$T_j = 2An_j - n_j^2 - 1, \qquad H_j = \sum_{i \in I_j} Y_i(m_i - X_i),$$

$$S_j = \sum_{i \in I_j} ((m_i - X_i + Z_i)^2 - Y_i^2), \qquad E_j = S_j - 2n_j H_j.$$

Delete the four private old pool coordinates and four rows, introducing one unrestricted integer ζ and the single residual

$$E_0 T_1 + E_1 T_0 - \zeta T_0 T_1 = 0. \tag{20}$$

It has degree at most five. All other residuals remain quadratic.

Theorem 11.1. *Under the retained canonical-Pell and square-frame hypotheses,* (20) *is solvable in $\zeta \in \mathbb{Z}$ if and only if all thirteen outputs are the exact powers $m_i = n_j^{k_i}$. An index is nonnegative whenever its base has absolute value at least two; unit bases retain their canonical signed powers. No coprimality of T_0, T_1 is required.*

Proof. Write $\delta_i = m_i - X_i + (A - n_j)Y_i$ for $i \in I_j$. Exact expansion gives

$$\sum_{i \in I_j} \delta_i^2 = E_j - T_j \sum_{i \in I_j} Y_i^2.$$

Thus (20) says that $\sum_{j,i} \delta_i^2 / T_j$ is an integer. Let $b = \lfloor \sqrt{W-1} \rfloor \geq 16$ and $C_* = [q(q+1)]^b \neq 0$. Every signed index satisfies $|k_i| \leq b$. The integer

$$d_i = C_* m_i - (C_*/n_j^b) n_j^{b+k_i}$$

is defined without a negative exponent. Canonical Pell congruences imply $C_* \delta_i \equiv d_i \pmod{T_j}$, because n_j is a unit modulo its own $T_j \equiv -1 \pmod{n_j}$. Therefore

$$\sum_{j=0}^{1} \sum_{i \in I_j} \frac{d_i^2}{T_j} \in \mathbb{Z}.$$

We only multiplied by C_*^2 and changed a numerator by a multiple of its own denominator. Neither C_* nor the other modulus was inverted.

The frame gives $|n_j|, |m_i|, |k_i| \leq b$. Hence

$$|C_*| \leq b^{2b}, \quad |C_*/n_j^b| \leq b^b, \quad |d_i| \leq b^{2b+1} + b^{3b} \leq 2b^{3b}.$$

With $r = 13 \leq 15$,

$$\sum_i d_i^2 \leq 4rb^{6b} < W^{3b+1} \leq W^W < \min(|T_0|, |T_1|).$$

Indeed $4r < W$, $b^2 < W$, $3b + 1 \leq b^2 + 1 \leq W$, and $|T_j| \geq 2|A| - W > W^W$. Also $\operatorname{sgn}(T_j) = \operatorname{sgn}(An_j)$, so the denominators have the same sign. The displayed integral reciprocal sum has absolute value below one and summands of a common sign. It is zero and every $d_i = 0$. Dividing the identity in $\mathbb{Q}$ by the nonzero C_* gives $m_i = n_j^{k_i}$. A negative index at a nonunit base would force an integer times a positive power of its absolute value to equal one, impossible.

Conversely exact powers give $T_j \mid \delta_i$ for every request. The square identity gives $T_j \mid E_j$, and $\zeta = E_0/T_0 + E_1/T_1$ is an integral solution of (20). □

The common sign and strict bound are essential. With opposite denominators, nonzero squares can cancel; without the bound a nonzero reciprocal sum can be an integer. The proof uses neither phenomenon.

11.3 Restoration and the consistency equivalence

At a new zero, first the unchanged signed cap and orientation recover the canonical pairs. The affine radix chart already gives two nonzero bases, so the theorem applies without invoking an unauthenticated radix power. Restore the old coordinates by

$$L_j = E_j/T_j, \qquad K_j = 2H_j + (2A - n_j)L_j.$$

They are integral on this zero set, and both old rows $K_j - 2H_j - (2A - n_j)L_j = 0$ and $S_j - n_j K_j + L_j = 0$ vanish. Only now apply the accepted old source bootstrap. Its retained singleton BH request, nonzero even H, odd nonzero B and endpoint argument exclude $x = 0$ and hence $q = 1$. The recovered q powers then give $y > 0$ and the large endpoint bound on x, excluding $q + 1 = -1$. Thus both bases are nonunits at every actual zero, the representative signs are authenticated in the old acyclic order, and a finite syntactic refutation is decoded. The reciprocal step itself did not assume these later conclusions. Conversely the cofinal construction above supplies a source zero in the restricted chart, and $\zeta = L_0 + L_1$ supplies the new zero. These are witness-existence maps, not a polynomial automorphism of the whole ambient lattice.

The finite compiler checks that the four deleted coordinates are private to the four protected rows, retains every singleton request and every norm/orientation/cap equation, and substitutes $q = 3z + 1$ throughout the entire retained source and every actual frame atom. Protected transported operations and two integral radical projections are rechecked against that source. The literal degree-five row and its complete squared expansion determine the stated exact degree 10 and coefficients. This separate composition proves the same consistency equivalence on $\mathbb{Z}^{212}$ without changing the principal quartic or claiming a global variable minimum.

12 Unit homogenized equality at coefficient height one

The source compiler acts on a finite quadratic system over unrestricted integers. The inherited Round 20 construction already supplies certified normalization, opposite-occurrence fusion, polarization, anchored equality and internal-flow contraction. We give its complete energy and witness argument to establish the interface, then prove the new monic product and chained-scale constructors. Unit-homogenized cycles are retained from Round21. Every height claim concerns the entire collected polynomial, including all equality penalties.

12.1 Certified normalization and row orientations

Let the given source be $R_r(x) = 0$. We first use the preceding exact affine refactorization, when applicable: if the homogeneous quadratic part of a row is $\eta L(x)^2$, with $\eta = \pm 1$, introduce $z = L(x) + k$ and replace the row by $\eta z^2 + R_r(x) - \eta(L(x) + k)^2 = 0$. The new definition is monic and its value is uniquely determined. Shared affine definitions can be reused. Coefficient-by-coefficient substitution verifies every transformed row. The provided original source, transformed source and factor certificate fix this operation completely.

Normalize integer coefficients by monic definitions of integer constants $t_c = c$ and scale variables $s_{c,i} = cX_i$. A constant is defined from already defined smaller constants as $t_c = t_a + t_b$ when $a + b = c$, using $t_1 = 1$. If no such pair is available, introduce the smaller binary intermediate and use $t_c = t_{\lfloor c/2 \rfloor} + t_{\lfloor c/2 \rfloor} + (c \bmod 2)$. Each added definition is triangular and monic. The delivered DAG prefers two already-needed constants; in particular it does not introduce a binary intermediate merely because a binary-only recipe would do so.

For each absolute quadratic coefficient $c > 1$, form a finite graph on the source coordinates, with an edge $\{i, j\}$ for every term cX_iX_j. A loop at i requires selecting i. Choose any vertex cover V_c of this graph, and introduce $s_{c,i} = t_c X_i$ only for $i \in V_c$. A term cX_iX_j is then represented by $s_{c,i}X_j$ if i was selected, or by $X_i s_{c,j}$ otherwise. The cover condition proves that one option always exists. The retained deterministic greedy cover starts with the loop vertices and repeatedly selects

a highest-degree remaining vertex; optimality is not assumed. Constants and linear terms use t_c and $t_c X_i$, respectively. Unit coefficients need no constructor.

Splitting repeated summands into separate occurrences yields

$$S_r(y) = \sum_j \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every normalized coordinate has a unique integral value over an original tuple. The checker verifies all triangular pivots and each exact source-row pullback, rather than trusting the constant-chain or cover-selection recipe. Thus normalization preserves both witness maps. The coefficient-factor choice, just like the later graph, is fixed by the certificate and is not a parameter of the polynomial.

Choose a fixed sign $\sigma_r \in \{-1, 1\}$ for each normalized row, and replace S_r by $T_r = \sigma_r S_r$. This preserves both its zero set and uniqueness of all constructors. The choice is part of the finite certificate, not a parameter of the final polynomial. The supplied heuristic flips a row when doing so reduces

$$\sum_m \left| \sum_{r,j:m_{rj}=m} \sigma_r \epsilon_{rj} \right|.$$

This balances opposite occurrences of the same logical monomial. Correctness does not require optimality of this heuristic.

12.2 Share a coefficient across both monomial factors

The retained split-factor method supplies a further exact normalization step. For a quadratic occurrence cxy, choose positive integers a, b with $|c| = ab$. Let $S(1, x) = x$; for $d > 1$ introduce a shared coordinate $S(d, x)$ with monic definition $S(d, x) - K_d x = 0$, where $K_d = d$ is fixed by the retained triangular constant chain. Replace the occurrence by

$$\operatorname{sgn}(c) S(a, x) S(b, y).$$

It remains a degree-at-most-two monomial of coefficient ± 1. Its unique constructor substitution is exactly cxy. If $x = y$ the same scale coordinate may occur twice, giving its ordinary square. Repeated definitions are shared; no unused scale coordinate is introduced.

This is an unconditional integral source equivalence. Given a source tuple, its monic constants and scales have unique values and every normalized row pulls back to its original row. Conversely those monic rows force the values and recover every original equation. There is no variable division or hidden divisibility assumption. The positive integer factor choices are frozen certificate data, not equation parameters.

The finite selection enumerates divisor pairs in both orientations and accepts a reduction of the number of used scales plus directly needed constants, then builds the actual constant chain. That cost is only a search heuristic. Exact final source pullbacks and coefficient expansion, not a presumed optimum of the heuristic, certify the delivered counts. The changed normalization alters the eventual graph namespace; its bijection with the old graph goes through the common arithmetic source, not an identity on arbitrary final tuples. The graph-energy and universal lift proof below applies unchanged to either normalized source.

12.3 Direct edges for cancelling occurrences

Create disjoint vertex blocks B_r for the rows and a hub block B_0. Inside each block choose a tree, with one independent unrestricted integer flow variable on each tree edge. A normalized occurrence is assigned to a vertex in its own row block. Partition the occurrences into sets of size one or two. A pair is allowed only when its occurrences are in distinct rows, have the same logical monomial, and have opposite signs in T_r.

A singleton of sign ϵ is represented by an edge from its row vertex to a hub vertex carrying ϵM. A pair $+m$ in row r and $-m$ in row s is represented by one edge from its row-r vertex to its

row-s vertex, carrying M. Here M is a degree-two homogeneous monomial in aliases of the logical factors of m: a linear term is multiplied by a unit alias and a constant uses two unit aliases. Each factor alias has one declared logical value. The two occurrences of a pair use *the same* edge and the same factor aliases.

The graph is required to be simple. The constructor rejects a pairing if its proposed unordered vertex pair was already used; remaining hub edges are placed without parallel edges. The certificate lists every bound source occurrence, and the checker verifies that each appears exactly once. More precisely, the edge boundary at the row-block level equals its list of signed source occurrences. Consequently, after substituting the intended logical values,

$$\sum_{v\in B_r} b_v = T_r(y), \qquad \sum_{v\in B_0} b_v = -\sum_r T_r(y). \tag{21}$$

where b_v is the divergence from prescribed edges alone. A fused edge contributes $+M$ and $-M$ to precisely its two bound rows; hence (21) is unchanged by fusion. This is the inherited occurrence-boundary identity.

12.4 The coefficient-height-one energy

Let d_v denote the outgoing-minus-incoming sum of all prescribed and free edges, and put

$$E = \frac{1}{2}\sum_v d_v^2.$$

Each edge has two endpoints, so its diagonal energy coefficient is one. Two distinct edges of a simple graph meet at most once, so their formal cross coefficient is 0 or ± 1. Give each prescribed product an initial private factor alias, and share its other logical factor as in the previous incidence compiler. The initial fully expanded energy has height one. Subsequently identify only aliases of the same logical value, and accept an identification only when full exact coefficient collection still has coefficients in $\{-1, 0, 1\}$. Such substitution preserves the sum-of-squares identity even when signed collisions cancel.

For every remaining non-unit logical class with at least three aliases use

$$C(a_1, \dots, a_k) = \sum_i a_i^2 - \sum_i a_i a_{i+1}, \quad a_{k+1} = a_1.$$

Since $2C = \sum_i (a_i - a_{i+1})^2$, it is nonnegative and vanishes exactly when all aliases agree. A singleton needs no penalty. A class with two aliases is either completed to a three-cycle by one dummy, or uses

$$Q(a, b) = a^4 + b^4 - a^3 b - ab^3 = (a - b)^2(a^2 + ab + b^2).$$

This is nonnegative and vanishes exactly at $a = b$; also $2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2$. The latter option is used only when neither alias occurs as a prescribed square, which ensures its four monomials cannot collide with the incidence energy. For all unit aliases, adding dummies if necessary so that $\ell \geq 2$, put

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U is $(u_1 - 1)^2 + \sum_{i<\ell}(u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus the final literal polynomial is

$$\boxed{P = E + \sum C + \sum Q + U.} \tag{22}$$

It is nonnegative over the reals. All monomials have total degree at most four. The full coefficient files are independently checked to have exact degree four, every nonzero coefficient equal to $+1$ or -1, constant coefficient one, and every stated variable present. The retained square-alias transformation below preserves this energy identity and revises a selected equality penalty with its exact zero set intact.

12.5 Forward map, inverse map, and uniqueness

Given a source tuple, evaluate the monic affine, constant and scale constructors, assign each ordinary alias its logical value and every unit alias the value one. Assign equality dummies the value of their class. All equality and unit penalties then vanish.

Root each internal tree at its first listed vertex. For a tree edge with child v, let $\tau_v = \pm 1$ be its incidence sign at the child, and set

$$w_v = -\tau_v \sum_{t \text{ in the subtree of } v} b_t.$$

This is an explicit integral polynomial in the source coordinates. Eliminating leaves proves that all nonroot divergences vanish. Formula (21) gives root divergence T_r in each row block and $-\sum_r T_r$ in the hub block. Thus the canonical lift satisfies, universally,

$$P(L(x)) = \frac{1}{2}\left(\sum_r R_r(x)^2 + \left(\sum_r \delta_r R_r(x)\right)^2\right). \tag{23}$$

where δ_r is the certified sign of the normalized row descending from the original residual R_r. Constructor rows pull back to zero. In particular every source zero has an integer lift. The affine-definition rows also vanish identically under their canonical substitution, so (23) composes unchanged through affine refactorization. One must not replace (23) by a bare sum of the original squares.

Conversely, if $P(z) = 0$, nonnegativity in (22) forces every equality penalty and U to vanish. Hence all ordinary aliases in a class agree, and all unit aliases equal one. Also $E = 0$ forces every vertex divergence to vanish. Summing them over a row block cancels its internal tree flows; by the certified occurrence-boundary identity the result is precisely $T_r(y) = 0$. Since $\sigma_r = \pm 1$, all normalized rows $S_r(y)$ vanish. Triangular monic constructors then force the constant, scale and affine coordinates to their intended values. Reading the original logical coordinates yields an original source zero.

The same argument proves uniqueness: constructors, aliases, units and dummies have unique values over the original tuple, and leaf elimination uniquely determines every internal tree flow. Thus the displayed forward map and coordinate projection are inverse bijections between integer zero sets. No sign restriction on a final variable, division by a variable, or unproved boundedness assertion is used.

12.6 Square polarization removes equality dummies

Suppose a logical value currently has exactly two factor aliases a, b, and its equality penalty uses a fresh dummy d:

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

The reason for retaining this dummy can be a prescribed square a^2 inside the incidence graph: directly substituting the two-alias penalty $Q(a, b)$ may collide with its degree-four diagonal contribution. The retained Round 19 operation first changes selected prescribed squares into products of same-logical aliases, $a^2 \mapsto ab$ (or $b^2 \mapsto ab$). It then replaces $C(a, b, d)$ by $Q(a, b)$ and deletes the dummy coordinate d.

Theorem 12.1. *Make any finite selection of such same-logical prescribed-square changes, and replace each affected two-alias equality cycle by Q. If the full collected final polynomial has coefficient height one, it is a height-one quartic with an integer zero set bijective to the original arithmetic source. Each deleted equality dummy reduces its dimension by one.*

Proof. Changing a prescribed edge's label from a^2 to ab leaves its two incidence endpoints and orientation unchanged. The graph energy therefore remains exactly $\frac{1}{2}\sum_v d_v^2$, with its divergences

evaluated using the modified labels. It is still nonnegative, and all labels still have degree at most two. Also

$$Q(a,b) = (a-b)^2(a^2+ab+b^2) \geq 0,$$

and $Q(a,b) = 0$ if and only if $a = b$: the second factor equals $(a+b/2)^2 + 3b^2/4$ and vanishes only at $a = b = 0$.

At any zero of the final energy, every nonnegative equality term is zero, so same-logical aliases agree and units equal one. Consequently every polarized label ab has exactly the value previously specified by a^2. The occurrence-boundary identity is unchanged after logical substitution. Vanishing vertex divergences therefore give all normalized source rows, and monic reconstruction gives an actual arithmetic source zero.

Conversely, for an original source zero assign both a and b their common logical value. All polarized edge labels then have their old values, all new Q terms vanish, and the same unique subtree flow assignment makes every vertex divergence zero. Deleted dummies have unique old value $d = a = b$, so extension and deletion are inverse on the zero sets. This also proves that the source-polynomial lift identity (23) survives universally, not only at source zeros.

The degree is at most four by construction. Exact degree four and coefficient height one are finite properties of the entire collected polynomial and are explicitly checked. They do not follow merely from the simplicity of the incidence graph after monomial substitution. The operation is accepted only after this full collection check, including all equality and unit terms. Thus possible collisions do not invalidate the claim. □

The theorem makes no assertion that an arbitrary polarization is safe for height one, nor that a chosen finite sequence is optimal. It separates the unconditional witness proof from the coefficient test selecting a particular literal endpoint.

12.7 Absorbing equality dummies into anchored products

The retained anchored operation removes an equality dummy without changing any prescribed square label. Suppose two aliases a, b of one non-unit logical value are tied by a fresh dummy d through

$$C(a,b,d) = a^2 + b^2 + d^2 - ab - bd - da.$$

Assume d occurs nowhere else. Let u be an existing unit alias already anchored to one by U. Substitute $d = au$ and delete the coordinate d. The new equality penalty is

$$\widetilde{C}(a,b,u) = a^2 + b^2 + a^2u^2 - ab - abu - a^2u. \tag{24}$$

It has degree four and is the literal polynomial $C(a,b,au)$.

Theorem 12.2. *Under the stated incidence restrictions, replacing $C(a,b,d)$ by (24) preserves the integer zero set by a polynomial bijection after deleting one coordinate. It also preserves the universal source lift identity (23). If the entire resulting collected polynomial has height one, exact degree four and all declared coordinates present, it is a certified height-one quartic with one fewer independent integer variable.*

Proof. For arbitrary real a, b, u,

$$2\widetilde{C} = (a-b)^2 + (b-au)^2 + (au-a)^2 \geq 0.$$

At a zero of the complete energy, its unchanged unit penalty forces $u = 1$. The new equality penalty then vanishes exactly when $a = b$. Thus the old dummy has the unique value $d = a = b$, while every other summand is unchanged. Extending with $d = a$ maps a new zero to an old zero. Conversely, at an old zero $d = a = b$ and $u = 1$, so deleting d gives a new zero. These maps are inverse polynomial maps on the zero sets.

For a canonical source lift, all unit aliases equal one and all ordinary aliases equal their logical values. Consequently $au = a = d$ identically on that lift, including away from source zeros. No prescribed incidence label or divergence has changed, so the universal lift identity is retained.

Polynomial substitution preserves nonnegativity but not automatically coefficient height. The term a^2u^2 may collide with graph-energy terms, and several absorbed dummies may share a unit alias. The acceptance test therefore collects the whole energy after every proposed substitution, including all other penalties. It explicitly checks the finite claims about coefficients, degree and the live coordinate set. No extra unit coordinate is introduced: u was already present and anchored. □

Several such absorptions can be composed, using the same or different existing unit aliases, provided the full polynomial test accepts the frozen final choices. Unlike Round 19 polarization, this operation leaves the prescribed square monomials in the incidence graph untouched. Unlike a formal substitution $d = a$ alone, it does not insert a coefficient -2 into the equality row before the complete energy is considered. Its quartic product with an already anchored unit provides additional distinct monomials that can retain height one.

12.8 Contracting an internal flow edge

The retained operation changes the divergence energy rather than only the alias penalties. It removes a flow coordinate on an internal tree edge and identifies its endpoints. It is therefore not an alias identification or an instance of square polarization. Its soundness is unconditional; coefficient height one is a separate exact acceptance test.

Theorem 12.3. *Let an incidence presentation have the energy form* (22)*, with every internal edge belonging to a tree inside one row block or the hub block. Choose an internal edge from u to v, with its own flow coordinate w. Delete that edge, identify its vertices, and replace their two divergences by their sum. Retain all prescribed occurrence labels and equality and unit penalties. Then the old and new integer zero sets are in polynomial bijection, preserving the original arithmetic coordinates. The new presentation has one fewer flow coordinate and degree at most four. If full collection of its polynomial has height one, exact degree four, and every listed variable occurs, it is a certified height-one quartic on that smaller coordinate set.*

Proof. Write the old two divergences as

$$d_u = a_u + \epsilon w, \qquad d_v = a_v - \epsilon w, \qquad \epsilon \in \{-1, 1\},$$

where a_u, a_v are integer polynomials independent of w. No other divergence or penalty depends on w, because it is the independent flow coordinate of this one edge. The merged divergence is $d_{uv} = a_u + a_v$. Thus the new polynomial is exactly

$$P' = \frac{1}{2}\left((a_u + a_v)^2 + \sum_{t \neq u,v} d_t^2\right) + \sum C + \sum Q + U.$$

This expression proves nonnegativity and the degree bound without any claim that the old and new polynomials themselves agree.

At an old zero, nonnegativity forces $d_u = d_v = 0$ and every other summand to vanish. Deleting w therefore gives $a_u + a_v = 0$ and a new zero. Conversely, at a new zero all its displayed summands vanish. Set

$$w = \epsilon a_v.$$

Then $d_v = 0$, while $d_u = a_u + a_v = 0$; all other old summands are unchanged. Indeed the stronger identity $P|_{w=\epsilon a_v} = P'$ holds for every remaining tuple: the two old divergences become $a_u + a_v$ and zero. This is an integral polynomial restoration, and it is unique on zeros because ϵ is a unit.

Restoration and deletion are mutually inverse on the zero sets. All equality and unit arguments, and hence all recovered logical coordinates, are identical on the two sides.

The edge's endpoints belong to one block. Summing the contracted vertex divergences in that block therefore gives the same boundary sum as before contraction. In particular (21) continues to recover exactly the same normalized source row; no source occurrence is deleted. Contracting an edge of a tree leaves a tree. The remaining flows are still uniquely recovered by leaf elimination, or equivalently by composing the one-edge restoration above.

Finally, degrees and coefficient heights concern the polynomial after all monomial collisions, including equality and unit terms. Contracting vertices can create parallel prescribed incidences, so the initial simple graph argument alone is no longer a sufficient height certificate. The selection test reconstructs and collects the entire new energy and checks its actual integer coefficients. This establishes the stated height, exact degree and coordinate census only for the accepted finite instance. □

Corollary 12.4. *Any finite certified sequence of accepted internal-edge contractions preserves the calibrated integer zero set. Its inverse is obtained by restoring the deleted flows in reverse order. If all k deleted flows were distinct live coordinates and no other coordinate is removed, its independent-variable count decreases by exactly k.*

Proof. Compose the polynomial deletion and restoration maps of Theorem 12.3. Every step preserves the block boundary identity and leaves internal blocks as trees. The final independently collected array fixes the actual coordinate count and all measured polynomial invariants. □

This argument does not predict which edges pass the height test or minimize the support. A contraction can save a variable while creating more monomials; it can also fail the height test entirely. Adaptive selection is a construction method, not a proof of optimality. The released certificate freezes each accepted edge, so reproduction requires no heuristic search.

12.9 Unit homogenized equality cycles

The retained Round21 support reduction generalizes the anchored three-cycle without adding a coordinate. Let a, b be the two aliases of one non-unit logical class, and let $u_1, \dots, u_\ell$ be existing unit aliases, already anchored by the unchanged unit energy U. Choose three expressions f_1, f_2, f_3, each equal to a, b, au_j or bu_j, with both underlying aliases a and b represented. Replace that class's equality penalty by

$$C(f_1, f_2, f_3) = f_1^2 + f_2^2 + f_3^2 - f_1 f_2 - f_2 f_3 - f_3 f_1. \tag{25}$$

No other incidence label or unit relation is changed.

Theorem 12.5. *The replacement preserves the calibrated integer zero set and the universal source lift. It introduces no variable and has degree at most four. Any frozen instance whose complete coefficient collection has height one and exact degree four is a height-one quartic on the same independent coordinate set.*

Proof. The polynomial is nonnegative for every real tuple because

$$2C(f_1, f_2, f_3) = (f_1 - f_2)^2 + (f_2 - f_3)^2 + (f_3 - f_1)^2.$$

At a zero of the complete energy, the separately nonnegative unit summand gives $u_j = 1$ for every unit alias. The three expressions then reduce to copies of a and b, with both present. Their equality is therefore exactly $a = b$. Conversely $a = b$ and all $u_j = 1$ make the new penalty vanish. All other summands and the source-block divergence identity are unchanged. The old and new energies have literally the same zero set in the same coordinate space, so this step has the identity

map as both witness maps. The same source projection and unique tree-flow lift also remain inverse on zeros.

On the canonical lift of an arbitrary source tuple, all units are one and all aliases equal their logical values. Thus the new penalty vanishes identically there, not merely over source zeros. This proves that (23) is unchanged. Each f_i has degree at most two, so the new penalty has degree at most four.

These facts do not by themselves prove height one or decreased support. A product au_j can create monomials already present elsewhere. The complete energy is therefore collected after each proposed change, including incidence, other equality penalties and the unit path. The accepted frozen certificate must pass exact checks of every coefficient, degree, constant and live coordinate. No choice of f_i is assumed universally safe for height. □

This changes the polynomial away from its zeros, but retains the same source witnesses. The tested cycles save support through actual signed monomial collisions in the full energy, not through an uncollected count of terms in one penalty. The finite selection procedure is not an exhaustive search over height-one equations.

12.10 Product constructors for constants

The new normalization retains the old logical coordinate list, constant values, scale values and source rows. If a needed constant K_c has available earlier constants K_a, K_b with $1 < a \leq b < c$ and $ab = c$, replace its additive definition by

$$K_c - K_a K_b = 0.$$

The chosen pair is the lexicographically least (b, a) among the eligible pairs. This is fixed certificate data, not an unknown of the equation.

Proposition 12.6. *These replacements preserve the normalized integer zero set literally, and preserve the final network zero set by the canonical source lift and projection. They introduce no new normalization coordinate or sign branch.*

Proof. Constants are ordered increasingly. Both factors precede K_c, even when $a = b$. Induction through the unchanged additive and new multiplicative monic definitions forces $K_c = c$ uniquely. Conversely the intended values satisfy every definition. Every source row and scale has exactly the same pullback. Thus the normalized zero sets agree in the same coordinates. Each final network separately has the already proved inverse projection and unique lift through the common source, which supply the desired bijection even though the final aliases and flows may differ. The universal identity (23), including the hub term, remains valid. No division is used. Positivity of a forced constant is not a restriction on a quantified final coordinate. □

The monic row now has two signed monomial occurrences rather than three. This changes the graph search and can remove more than one eventual alias or flow; final savings are measured after complete coefficient collection.

12.11 Chained scale constructors

The smallest selected endpoint also replaces a scale definition $S(c, x) - K_c x = 0$ when an earlier scale $S(b, x)$ and a constant K_d exist with $d > 1$, $b < c$ and $db = c$. The new definition is

$$S(c, x) - K_d S(b, x) = 0.$$

Constants precede all scales and scales are ordered by (coefficient, source coordinate). Induction therefore forces $S(c, x) = d(bx) = cx$ for every integer x, including negative and zero values. Conversely those values satisfy the new rows. The monic pivot and predecessor on the *same*

original coordinate are essential; merely factoring the numerical coefficient without checking the base coordinate would not establish the identity. This again preserves both normalized zero sets and the final source-lift bijection without division. The deterministic choice is the lexicographically least available (d, b).

The independent checker removes row orientations and verifies the identical coordinate list and metadata, every changed and unchanged row, all earlier-factor conditions, every exact source pullback, and the complete collected final polynomial. It tests source/constant corruption classes and a separate chained-scale corruption. No heuristic claim of optimality is used to prove correctness or numerical metrics.

12.12 The selected height one catalog

All entries below have unrestricted integer coordinates, exact degree four, coefficient height one, constant one, and are nonnegative on all real tuples. Support counts the constant monomial.

Endpoint	Variables	Terms	Aliases	Flows
minimum493	493	24828	472	21
compact494	494	24648	473	21
compact496	496	20548	474	22
compact504	504	18297	478	26
compact507	507	17524	479	28
balanced546	546	12139	499	47
balanced557 retained	557	11991	510	47
sparse605	605	8305	510	95
sparser701	701	7141	510	191

All declared coordinates remain live; there are no equality dummies. The 493 and 494 profiles use the retained source241 family; the next three use source242, the two balanced points source263, and the last two source285. These names are source identifiers, not their refactored normalization dimensions. The original source bindings and their full proof closure are retained. None of these is silently rebased to the new CRT-anchor source.

The old 503/24183 and 505/20405 variable/support points are dominated by 496/20548 and 504/18297, respectively. The 512/18184 point is dominated by 507/17524, and 617/8314, 713/7172 by 605/8305, 701/7141. The old 557/11991 remains a genuine selected tradeoff against 546/12139. The 493/24828 minimum is a dimension improvement, not a support improvement over 503/24183. These are comparisons within the finite stated catalog, not a global Pareto frontier.

13 Linear trace and inverse correction in ten positive integers

The three equations in this section have exactly ten independent inputs

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

Their complete four-port tuple is the Start-only4127 specialization of Section 14. The arithmetic proof below is uniform under its authenticated scalar bounds; the whole-native bridge in that section licenses all four ports together. Strict Horizons supplies the free quotient and projection-first strategy; Trace Radical supplies the linear trace; Unsquared Factors and Linear Selectors supplies the inverse correction at k. The retained Round21 uniform signed bound licenses their composition without assuming $q \mid Y$ early and without the extra positive shift $T + \alpha$.

13.1 The three complete polynomial definitions

Put $h = \beta J$, $q = h + 1$, $C = q - F - \alpha$ and $k = \eta + \zeta$. Choose one of the following fixed charts. The symbols W_b, λ and D below are polynomial abbreviations, not independent inputs.

Chart	W_b	X	Y	λ	D
Free plain	qw	q^2w	s	q	q^7
Scaled inverse	w	qw	qs	1	$6q^6$
Free inverse	w	qw	s	q	$6q^7$

In every chart define

$$\begin{aligned}
c &= kY+\eta, & A &= \lambda Y(X+1)+1, & \Delta &= A^2-1,\\
H &= 4A-5, & v &= X\lambda^2Y^2, & E &= 1+v(v+1)k^2,\\
R &= [h(q-F)+\alpha](q^2-1)+(M_C+qM_F)J,\\
B &= (K_0+W_b)C-F, & U &= ic^2, & a &= \Delta U^2E,\\
F_0 &= 1+a, & G &= EF_0, & f &= 1+2a,\\
Q &= 4\Delta aF_0, & b &= cT-R, & V &= fb-c,\\
K &= X(3c-2X)+2, & L &= bV, & M &= hDH.
\end{aligned}$$

In the plain chart put $e_c = k-R-1-q^4s$; in either inverse chart put $e_c = 6(k-R-1-q^4s) - vk(k^2-4)$. Finally set

$$\begin{aligned}
Z &= hD[H(n-LB)+K]-H[DB+he_c],\\
P &= Q(Z^2-M^2GV^2)+(M^2G-Z^2).
\end{aligned} \tag{26}$$

The equation is $P=0$. Fixed powers are implemented by binary multiplications. There is no radical or quotient gate. The graph's register sharing, not the typography of these abbreviations, defines the measured operation count.

For the proof only, let $t_* = B/h$, $\kappa = K/H$, $z = Z/M$ and $W_* = 1+hL$. In the plain chart let $e = k-R-1$, $\rho = e/q^7$. In the inverse charts let

$$e_* = k-R-1-\frac{vk(k^2-4)}{6}, \qquad \rho = e_*/q^d, \quad d = 6 \text{ or } 7.$$

In all charts put $\sigma = Y/q^3$. Direct expansion gives

$$P = (Q-1)Z^2 - M^2G(QV^2-1), \qquad z = n+\kappa+\sigma-W_*t_*-\rho. \tag{27}$$

These proof quotients are rational until integrality is established. The carrier denominator is nonzero for every positive input.

13.2 Root extraction before any transport sign

Positive inputs give $q \geq 32$, $X \geq q$, $Y \geq 1$, $k \geq 2$, $c \geq 3$, $A > q^2$, $\Delta > 1$, $U \geq c^2$ and $E > 1$. The exact identities are

$$f^2 = 1+4aF_0, \quad Q = \Delta(f^2-1) = 4\Delta^2U^2G, \quad \gcd(E,F_0) = 1. \tag{28}$$

Also $f > 2c$ and $Q > c^2$, hence $V \neq 0$ for every integral b. At a zero, valuations in (27) imply that $G = g^2$ for a positive integer g: every prime of G divides neither $Q-1$ nor QV^2-1, so its valuation in the rational square z^2 is even. Coprimality then gives $E = \tau^2$, $F_0 = f_0^2$ and $g = \tau f_0$ with positive integer roots.

The inherited adjacent-coefficient descent makes $y = |z|/g$ integral. For completeness, a reduced denominator m satisfies $m^2 \mid Q-1$. For $u = |V| > 1$, the norm-preserving transformation

$$u' = (2Q-1)u - 2(Q-1)y, \qquad y' = (2Q-1)y - 2Qu$$

has a nonzero integral first coordinate with $|u'| < u$ and preserves the reduced denominator of the second. The first integrality follows from $(Q-1)y \in \mathbb{Z}$; denominator preservation follows from

$\gcd(2Q-1,m)=1$. The norm gives $u<y<(2Q-1)u/[2(Q-1)]$, proving $0<u'<u$ after reflection. Descent reaches $u=y=1$, forcing $m=1$. Therefore

$$QV^2-(Q-1)y^2=1, \qquad z=\varepsilon gy, \quad \varepsilon\in\{-1,1\}.$$

With $I=2Ug$, equations (28) give $f^2-\Delta I^2=1$, $Q=(\Delta I)^2$, $c^2\mid I$ and $c\le I$. The retained signed all-index folding theorem applies to $V\equiv -c \pmod f$ and gives $c=\psi_A(p)$ for a positive odd index p. Since $c\ge 3$, $p\ge 3$, and hence

$$c\ge c_3:=4A^2-1=4\Delta+3>4q^4, \qquad c>X, \qquad K>0. \tag{29}$$

No sign of R,B,C,b,V or divisibility $q\mid Y$ was used.

13.3 A uniform bound removes the positive shift

If $b=0$, the auxiliary norm would give $0<y^2-c^2=(c^2-1)/(Q-1)<1$, impossible for integers. Thus $b\neq 0$, $\operatorname{sgn}V=\operatorname{sgn}b$ and $|V|>1$. Put $S=\Delta I$, so $Q=S^2$. Pell classification of (SV,y) gives an odd auxiliary index: even traces are units modulo S, whereas SV is divisible by S. Index one would give $|V|=1$. The index is therefore at least three, and $|V|\ge 4Q-3$. Moreover

$$\frac{a^2}{g^2}=\frac{\Delta U^2 a}{1+a}>\frac{\Delta U^2}{2}>9,$$

so $f>6g$. Using $c_3\le c$ and $f>2c$ yields the essential bound, for either sign of V,

$$|b|\ge\frac{|V|-c}{f}\ge 4\Delta f-\frac{c+c_3}{f}>4\Delta f-1>4g. \tag{30}$$

The possible auxiliary index three is harmless; it is not excluded. The norm also gives $y<3|V|/2$. Consequently

$$L=bV=|b||V|>0, \qquad gy<3L/8<L/2, \qquad |b|/q^d<L/2 \quad (d=6,7).$$

This remains valid at $Y=1$, before its cubic q-scale is recovered.

In the plain chart $e=k-R-1<b$ since $cT-k+1>0$. In the inverse charts $k\ge 2$ gives $e_*\le e<b$. Thus $\rho<|b|/q^d$ in all charts. If $B\le -1$, then $W_*t_*<-L$, and (27) implies

$$n=\varepsilon gy-\kappa-\sigma+W_*t_*+\rho<L/2-L+|b|/q^d<0,$$

contradicting the positive domain. Hence $B\ge 0$ and $C>0$. This handles all negative values of R and zero uniformly. Only now use the authentic whole-tuple mask bound $3q<R<q^4-2$. Equation (29) gives $c>2R$, hence $b,V>0$. The retained normalized strong-rank and signed auxiliary-index theorems apply to $V\equiv -c \pmod f$ and $V\equiv -R \pmod c$; the latter uses $f-1=2\Delta i^2c^4E$. With $c>2p,2R$ they yield $p=R$. Thus R is positive odd, and mask parity gives even q and odd J. Neither ρ nor σ has yet been assumed integral.

13.4 Authenticate the first index at every prime

The square $E=\tau^2$ gives $k=2\psi_{1+2v}(m)$ with $m\ge 1$. The case $m=1$ would give $k=2$, $\eta=\zeta=1$ and $c=2Y+1$ strictly between $\psi_A(1)=1$ and $\psi_A(2)=2A$. Thus $m\ge 2$. The first base $1+2v$ exceeds A, since $\lambda Y(2X\lambda Y-X-1)>0$. Monotonicity and $c>k$ then give $m<R$.

In the free plain chart $v=q^4ws^2$. Integral z and $\gcd(q,hH)=1$ give $q^7\mid e-q^4s$, hence $q^4\mid e$. The recurrence $k\equiv 2m \pmod{4v}$ shows $q^4\mid 2m-R-1$. Its absolute value is less than $R<q^4$; therefore $2m=R+1$.

In either inverse chart $v=q^3ws^2$. The inherited finite Lucas identity, valid at every index, is

$$k=2m+8vC_m+32v^2Z_m, \quad C_m=\binom{m+1}{3}, \quad Z_m\in\mathbb{N}.$$

Since k is even, $k(k^2-4)/6$ is integral: write $k=2u$ and use $3 \mid u(u-1)(u+1)$. Thus e_* is integral. Clearing the carrier and cancelling its literal factor six, not inverting six modulo q, gives $q^d \mid hH(e_* - q^4 s)$, hence $q^4 \mid e_*$. Put $t_0 = 2m$, $d_0 = 2m - R - 1$, $A_m = C_m + 4vZ_m$ and $S_m = k^2 + kt_0 + t_0^2 - 4$. Cubic subtraction gives exactly

$$3(e_* - d_0) = 4v^2(24Z_m - A_m S_m). \tag{31}$$

Indeed $t_0(t_0^2-4) = 48C_m$ and $k - t_0 = 8vA_m$. For any prime $p \mid q$, writing $a_p = v_p(q) \geq 1$, the identity gives

$$v_p(e_* - d_0) \geq 6a_p - v_p(3) \geq 4a_p.$$

This includes the prime three. Hence $q^4 \mid d_0$, and the strict size bound gives $d_0 = 0$. In all three charts,

$$m = (R+1)/2. \tag{32}$$

Full q^6- or q^7-divisibility of e_* has not been inferred here; the third-primary obstruction is repaired only after projection.

13.5 Projection precedes complete denominator recovery

The literal numerator and integral z give $H \mid hDK$. Cancel only the q-part: $H \mid d_1 hK$, where $d_1 = 1$ in the plain chart and $d_1 = 6$ in the inverse charts. Write $R = 2r+1$, so $r \geq 48$. The retained λ-uniform ratio theorem applies to the authenticated indices and strip $Y < c/k < Y+1$:

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X+1) < H.$$

Modulo odd H, the universal scaled recurrence gives $3 \cdot 2^{R-1}\psi_A(R) \equiv 4^R - 1$. The retained projector yields

$$H \mid d_1 h(X - 2^R)(2^R X + 1).$$

The complete product, including d_1, has magnitude below H:

$$d_1 h|(X-2^R)(2^R X+1)| < 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1} < H.$$

After division by X^{r+1} the two terms decrease for $X \geq 32$, $r \geq 48$; at $(32, 48)$ their sum is $2^{-117} + 2^{-25} < 1$. Thus $X = 2^R$. The universal projector itself then gives positive integral $\kappa = K/H$; no unproved cancellation of $d_1 h$ is used.

Now $q \mid X$ implies $q = 2^a$ with $a \geq 5$. Since $R > 3q \geq 6a$, one has $q^6 \mid X$. In the free charts $v = Xq^2Y^2$ is divisible by q^8. In the plain chart $4v \mid e$ gives $q^7 \mid e$; in the free inverse chart (31) with $d_0 = 0$ gives $q^7 \mid e_*$. In the scaled chart $Y = qs$, so $q^8 \mid v$ as well and the same identity gives $q^6 \mid e_*$. Thus ρ is integral everywhere.

The carrier now makes $\sigma - W_* B/h$ integral. Multiply by $q^3 h$ and reduce modulo q^3 to obtain $q^3 \mid Y$; reduce modulo h to obtain $h \mid B$, using $\gcd(q,h) = 1$ and $W_* \equiv 1 \pmod{h}$. Thus $t_* = B/h \in \mathbb{N}$. These are the exact retained native outer conditions. The fixed-minus parity and half-binomial theorems, uniform in λ, reconstruct fresh source witnesses at $A_{\text{source}} = Y(X+1)+2$, not at the present base. The retained whole-tuple native/refutation calibration then supplies a finite syntactic ZFC refutation.

13.6 A constructive positive lift with possibly negative correction

Conversely a finite refutation supplies canonical outer data $q = 2^a$, $X = 2^R$, $R > 3q$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, $C > 0$ and integral $t_* = B/h \geq 0$. Choose $w = X/q^2$ in the free plain chart and $w = X/q$ otherwise; take $s = Y$ in free charts and $s = Y/q$ in the scaled chart. All are positive integers. Construct fresh A, v and set

$$c = \psi_A(R), \quad k = 2\psi_{1+2v}((R+1)/2), \quad \tau = \chi_{1+2v}((R+1)/2), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The retained exact uniform ratio theorem gives $0 < \eta < k$. The same recurrence and power-of-two divisibility prove integral ρ; projection gives $0 < \kappa < Xc$, and σ is positive integral.

Choose a positive Pell matrix period at A modulo $c^2\tau$, obtaining a seed (f_0, I_0) with $c^2\tau \mid I_0$. Set $i = I_0/(c^2\tau) > 0$. Then $a = f_0^2 - 1$, $G = (\tau f_0)^2$ and the doubled strong pair is $f = 1 + 2a$, $I = 2f_0 I_0$. Keep this seed fixed. The retained positive auxiliary reconstruction gives (V_*, y) with $V_* = f^2(cT_{\text{old}} - R) - c$. Define

$$T_{\text{new}} = \frac{V_* + c + Rf}{cf}.$$

This is integral: $f^2 \mid V_* + c$, $V_* \equiv -R \pmod{c}$, $f \equiv 1 \pmod{c}$ and $\gcd(c, f) = 1$. It is strictly positive, and $V_* = f(cT_{\text{new}} - R) - c$ exactly. This is a fresh witness construction, not an inverse shear on arbitrary positive tuples.

In inverse charts the correction may be negative. Since $k \geq R + 1$ and $vk^2 > 6$, one has $|\rho| < vk^3/3$. For every chosen $i \geq 1$,

$$g > \sqrt{\Delta}c^2E > c^2v^2k^2 > vk^3, \qquad g > 4Xc.$$

For the latter inequality use $\sqrt{\Delta} > A - 1 > X$ and $cE > 4$. Thus $\kappa + \sigma < g/2$ and $|\rho| < g/3$. In the plain chart $\rho \geq 0$, and the same bound on g suffices. In all cases

$$n = gy + (1 + hL)t_* + \rho - \kappa - \sigma$$

is a strictly positive integer. Its carrier has $z = gy$, so the literal polynomial vanishes. Every one of the ten independent inputs has been constructed in the actual positive domain at one coherent tuple.

13.7 Attained degrees and exact circuit counts

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = k_0 sT - q_0^2(q_0 - F)$. Complete homogeneous collection gives

$$\begin{aligned}
[P_{\text{plain}}]_{240} &= -2^8 q_0^{112} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2, \\
[P_{\text{scaled}}]_{232} &= -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2, \\
[P_{\text{free}}]_{214} &= -9 \cdot 2^{10} q_0^{86} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2.
\end{aligned}$$

They have respectively $99, 202, 99$ collected monomials. In the free charts the leading part of b is $-q_0^3(q_0 - F)$; in the scaled chart the full degree tie $b_0 = q_0 B_0$ must be retained. Selecting $\eta^{32}F^2$ in the free cases gives nonzero coefficients $-2^8\beta^{112}$ and $-9 \cdot 2^{10}\beta^{86}$; selecting $\eta^{34}T^2$ in the scaled case gives $-9 \cdot 2^{10}\beta^{100}$. These prove attained degree at the authentic positive β.

In chart order, the degrees of (Q, M, G, V, Z) are $(82, 13, 52, 40, 60)$, $(80, 11, 50, 40, 57)$ and $(72, 12, 46, 36, 54)$. The unique dominant output term is $-QM^2GV^2$. The live binary-operation counts are respectively $82, 85, 87$. The independent checker verifies complete top forms of every register, not sampled lines. Historically, the scaled 232/85 point saved two operations against incoming Trace Radical's 232/87; the free 214/87 point saved eighteen degrees against that same incoming point. Neither comparison is a global bound or a claim about unrestricted integer variables.

14 A whole new coefficient tuple for the ten-positive family

The three ten-positive-variable equations retain their free-port arithmetic and their exact degree/gate pairs $(240, 82)$, $(232, 85)$, and $(214, 87)$. The new result is a coherent specialization of *all four* fixed ports to the Start-only complemented native encoding with cell length 4127. The complete unsigned coefficient payloads shrink from 295 388 to 97 046 bytes, a saving of 198 342 bytes (67.146%). The incoming Start-marker package had specialized its fifteen-positive P66 front to this tuple, but retained the old radix-67 tuple for its ten-positive 232/85 equation. The present result proves the missing whole-native bridge for all three retained ten-positive charts. It does not lower their degrees or operation counts, or change the separate six-port integer11 family.

14.1 The complete fixed tuple and its semantic requirements

Put $\rho = 2^{47}$, $M = 2^{44}$, $L = 4127$, $B_0 = \rho^L$ and $\beta = B_0 - 1 = 2^{193969} - 1$. Let $t_j = (w_j, n_j, e_j, s_j)$, $0 \le j \le 942$, be the complete fixed tile list reconstructed from the retained 436-character machine numeral. Its Start is $(2, 435, 3, 1)$ and tile942 is $(4, 437, 2, 437)$. Define

$$c_j = 1 + 2^{10}(1 + w_j + 2^8 n_j + 2^{17} e_j + 2^{25} s_j), \quad 0 \le j \le 942,$$

and, in addition,

$$\begin{aligned} c_{1002+j} &= 2^{10+j}, \quad c_{992+j} = 2^{27+j} \quad (0 \le j < 8), \\ c_{973+j} &= 2^{18+j}, \quad c_{943+j} = 2^{35+j} \quad (0 \le j < 9). \end{aligned}$$

All other c_e, $0 \le e \le 3296$, are zero. Write $G_m(x) = \sum_{j=0}^{m-1} x^j$ and

$$\begin{aligned} \Psi(x,t) &= x^{992}(x^{10} + t^{17})G_8(tx) + t^8 x^{943}(x^{30} + t^{17})G_9(tx) + t^{34}(2 - t)x^{1031}, \\ \Phi(x) &= \sum_{j=0}^{942} c_j x^j + 2^{10}\Psi(x, 2), \qquad \Lambda = \rho^{2042} + \rho^{3074}, \\ D_C &= \rho^{20} + \rho^{30} + \rho^{1031} + \Lambda\Phi(1/\rho), \\ K_0 &= D_C + B_0\rho^{10}, \\ M_C &= \beta + \rho - (\rho + 3)\rho^{990} - G_{943}(\rho) - \Psi(\rho, 1), \\ M_F &= B_0 + 3 + (M - 2)\Lambda + \rho^{973}G_9(\rho) + \rho^{1012}G_8(\rho) + \rho^{1031} + 2^{29}\rho^{4124} - \rho^{4106}. \end{aligned}$$

These finite sums define integers: the reflected exponents are nonnegative. M_F is already the once-shifted source mask; the proof-only native mask is $M_F - \beta$, not a second independently selected coefficient. The complete numerical payloads, decoder, all 943 tiles and 3297 weights are included, rather than replacing coefficients by hashes.

The retained Start-only proof establishes more than scalar inequalities. Its physical support has 980 positions, with a separate zero-weight flag at 1031. The carry box is $0, \ldots, 29$; clean clauses precede alignment. Eight Start label bits and the flag test leave offsets 0 and 32, and the actual complemented south color $511 - 437 = 74$ of tile942 excludes 32. The flag at a temporal predecessor equals the current Start selector, *not* occupancy. Positive complemented colors propagate occupation after alignment. Complete clause resets, origin masks and 875 padding bits then reconstruct the genuine marked blank computation. Every one of these semantic conditions is retained; matching port names or the inequalities $0 < M_C < \beta < M_F < 2\beta$ would not alone license this specialization.

For the converse the actual new controller has

$$d = 193969, \quad \lambda = 474491, \quad \mathrm{ord}_{47}(B_0) = 23, \quad \mathrm{ord}_{4127}(B_0) = 2063, \quad \mathrm{lcm}(23, 2063) = 47449.$$

The values of $1 + B_0K_0 + B_0^2$ modulo $47, 4127, 474491$ are respectively $9, 3487, 153928$, and $B_0^{\lambda-1} = 1 + 135285\lambda \pmod{\lambda^2}$. Full trial division and all prime-factor order tests discharge these finite premises. The whole cofinal rectangle/subset-sum proof uses this new K_0, not old controller residues or an old packed word.

14.2 Why all three arithmetic charts transfer

Write $h = q - 1$, $B_{\mathrm{tr}} = (K_0 + W)C - F$ and $t = B_{\mathrm{tr}}/h$. The retained ten-positive proof, with the new authenticated scalar boxes, recovers $X = 2^R$, $q^3 \mid Y$, $C > 0$ and integral $t \ge 0$. In the plain free chart $X = q^2 w$ and $W = qw$; in both inverse charts $X = qw$ and $W = w$. Thus in *every* chart $W = X/q$. Define fresh source coordinates

$$w_s = X/q, \qquad s_s = Y/q^3, \qquad z_s = t + 1.$$

They are positive integers and satisfy the exact transport, not merely a congruence:

$$(K_0 + w_s)C + (q - F) - z_s h = ht + q - (t+1)h = 1.$$

The rank $R = [h(q-F)+\alpha](q^2-1) + (M_C + qM_F)J$ and $C = q - F - \alpha$ agree literally with the new native interface. Projection still precedes full denominator recovery; no early inference from $q^3 \mid HY$ to $q^3 \mid Y$ is used.

For an explicit soundness bridge set $r = (R-1)/2$ and rebuild the unscaled source base, rather than identifying it with a free-chart base:

$$\begin{aligned}
&a_s = Y(X+1), && A_s = a_s + 2, && \Delta_s = A_s^2 - 1, && H_s = 4a_s + 3,\\
&c_s = \psi_{A_s}(R), && D_s = \chi_{A_s}(R), && P_0 = 1 + 2XY^2,\\
&k_s = 2\psi_{P_0}(r+1), && \tau_s = \chi_{P_0}(r+1),\\
&\eta_s = c_s - k_s Y, && \zeta_s = k_s - \eta_s,\\
&h_s = (k_s - R - 1)/(XY), && \gamma_s = (D_s - X - a_s c_s)/H_s.
\end{aligned}$$

The retained fixed-minus parity and exact half-binomial theorem prove $R = 3 \pmod 4$ and positivity and integrality of these witnesses. Take $m_s = c_s R$, $I_s = \psi_{A_s}(m_s)$, $f_s = \chi_{A_s}(m_s)$ and $i_s = I_s/c_s^2$. The Pell multiplication identity gives $c_s^2 \mid I_s$. The canonical auxiliary construction at $S_s = \Delta_s I_s$ supplies positive V_*, y_s, T_s with

$$V_* = f_s^2(c_s T_s - R) - c_s, \qquad S_s^2(V_*^2 - y_s^2) + y_s^2 = 1.$$

These fresh witnesses give every normalized P66 factor equal to1, at the *same complete new tuple*. The Start-only P66 soundness theorem therefore extracts a finite syntactic ZFC refutation. This establishes a semantic composition, not equality with the old specialized polynomial.

Conversely, the new native theorem constructs fresh canonical data from any finite refutation, with $q = 2^{dN}$, $X = 2^R$, $R > q(q^2-1)$, $R = 3 \pmod 4$, $\mathrm{pc}(R) = 3dN + 2$, $q^3 \mid Y$ and the actual integral transport. The rank margin permits $w = X/q^2$ in the plain free chart and $w = X/q$ in the inverse charts; take $s = Y$ for free charts and $s = Y/q$ for the scaled chart. The retained uniform ratio theorem, divisible strong period, trace-linear auxiliary reconstruction and positive-n bounds then produce all ten positive coordinates. The inverse correction may be negative; the original g bounds explicitly handle this. Thus each new polynomial has a positive zero exactly when a finite ZFC refutation exists. Negation is equivalent in ZFC to syntactic Con(ZFC), without deciding either side.

14.3 Exact degree and verification

With $q_0 = (2^{193969} - 1)J$, $k_0 = \eta + \zeta$ and $B_* = k_0 sT - q_0^2(q_0 - F)$, the full highest forms remain

$$\begin{aligned}
[P_{240}]_{240} &= -2^8 q_0^{112} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2,\\
[P_{232}]_{232} &= -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_*^2,\\
[P_{214}]_{214} &= -9 \cdot 2^{10} q_0^{86} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2.
\end{aligned}$$

Their selected nonzero coefficients establish actual degrees at these fixed numerals. Every original free-port gate is unchanged. Independent reconstruction decodes the complete machine numeral, all tiles and weights, and all coefficient bytes without importing the producer; it also checks full native geometry and controller premises. Normal and optimized-Python deep checks reject all 508 single-operator corruptions, and clean-copy replay reconstructs graphs and receipts byte-identically. These finite checks authenticate the artifacts; they are not an unbounded native/refutation proof or a new formal-kernel certificate.

The squared-transport variant is not silently transferred to the two free charts: their pre-rank auxiliary index-three case is allowed, and the proven bound relative to bV does not imply the required bound relative to b^2. Likewise no assertion of global minimality or a new signed11 degree floor is made.

15 Provenance and distinct comparison baselines

The repository comparison is pinned to

`ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b`.

The complete intake records distinguish this current repository snapshot from the earlier immutable commits inside each inherited proof package. An upstream source's own commit pin is not silently rewritten when a new comparison snapshot is inspected.

Three sources of improvement must be kept separate.

1. The incoming CRT-anchor package supplies reuse of the encoded row index and the range-free label proof. This round supplies the actual weighted/quadratic growth variants and their transported exact charts. The two-variable reduction against the Round21 principal is real, but it does not erase the incoming anchor's different sparse/high-height tradeoff.

2. The height-one branch keeps its original four arithmetic source families. Monic products and chained scales change the normalization and hence the graph, while preserving the common source zero set. Its smallest count493 is not obtained by compiling the new direct215 source. The retained557 point is identified as inherited, not new.

3. Start-only4127 supplies the incoming shortened native coefficient encoding. This round proves the previously missing whole-tuple transfer to all three ten-positive charts. Its comparison is against their actual old 295388-byte four-port tuple, not against a differently sized P66 tuple. The incoming P66 proof in the appendix makes its own separate 6007-cell comparison; those byte savings must not be added together.

The consolidated sources 166,168,170 and171 are comparison inputs, not new results of this round. Independent full coefficient recollection gives these additional points, all on unrestricted integers:

Source and chart	Variables	Degree	Terms	Height
166 balanced	230	4	4482	7968
166 compact	229	4	12006	7968
168 reciprocal	215	10	99914	113304
170 quotient carriers	217	4	19112	5184
171 separated radicals	217	4	25949	4576

These are not uniformly dominated by the new principal: for example the balanced230 point is substantially sparser, while the217 quotient-carrier point has lower height. The degree-ten reciprocal point is not another quartic. The source171 separated-radical route likewise remains a distinct comparison architecture, not an implicitly transferred scalar theorem. Its202-variable natural-domain point has 9818 terms and height 239204765; that is not a 202-variable unrestricted-integer equation. The intake authenticates 890 consolidated source blobs and independently recollects 117 incoming literal polynomials. Those finite checks do not make the four consolidated subtrees standalone-complete: their manifests still refer to inherited paths outside their respective subtrees. Their independent arithmetic architectures cannot be composed with the selected fronts merely by matching a coordinate count or a port name. Exact current source metrics and pins are retained in the release provenance record.

The separately audited compact graphs have the following source-declared domains and exact finite circuit metrics:

Source and chart	Domain	Variables	Degree	Gates
170 fixed residue	$\mathbb{Z}_{>0}$	13	206	70
170 fixed residue	$\mathbb{Z}_{>0}$	13	162	71
170 fixed residue q4	$\mathbb{Z}_{>0}$	13	154	72
170 rational carrier	$\mathbb{Z}_{>0}$	12	438	74
171 separated radicals	$\mathbb{Z}$	12	512	132

These degrees are independently attained at the authentic numerals; operation counts include all live binary gates. The finite degree checks are not an independent proof of each source's unbounded zero-set theorem. In particular the 13-positive degree154 point has lower degree than the new 10-positive degree214 point but more independent variables. Neither metric may be omitted when comparing the two architectures.

The primary comparison is multidimensional: domain, variable count, degree, collected support, coefficient height and live graph cost. Changing the domain from positive to unrestricted integers requires a proof, not a relabeling. An expanded direct quartic and a high-degree scalar circuit have different size regimes. The report makes no literature-priority claim or theorem that one endpoint is globally simplest.

16 Reproduction and evidence boundaries

The complete release is supplied in ordinary disjoint ZIP parts, each within the requested 4 MB limit, sharing one top-level directory. Extract all parts into the same parent directory before running the verifier. No individual part is advertised as a complete source closure.

The normative mathematical definitions are the complete finite residual, coefficient and graph arrays, not PDF timestamps. Every selected direct endpoint contains its ordered variables, all residuals and the fully collected polynomial. Every selected height-one endpoint contains its full coefficient array and semantic graph certificate. The scalar endpoints contain their complete acyclic graphs, all four actual coefficient words, the finite language decoder and all native-interface data. A digest is an identity check, never a substitute for a missing coefficient.

16.1 Component commands and report rebuild

From the combined extracted root, the principal finite checks are

```
python3 -B signed/build.py -check
python3 -B signed/verify.py
python3 -B reciprocal/verify.py -deep
python3 -B height_one/verify.py
python3 -B scalar/positive10/build.py -check
python3 -B scalar/positive10/verify.py -deep
```

The release README gives the integrated verifier and exact final command plan. Its receipt records terminal outcomes and source hashes. An integrity-only pass verifies bytes, not mathematical reconstruction. The Python finite checks use the standard library and do not require a network or a proof-assistant installation; the full inherited arithmetic and logical statements remain separate unbounded proofs.

Rebuild the native report with

```
sh report/build_report.sh
```

The script uses pdfLaTeX in a fresh temporary directory, compiles three passes, and rejects unresolved references and overfull boxes. The principal appendix is generated directly from its literal residual array by `report/render_residuals.py`; `RESIDUAL_BINDING.json` binds its source and generated TeX hashes. Independent re-parsing checks the complete displayed residual list, coefficient by coefficient. All final pages are rendered and visually inspected; the QA receipt records the exact PDF hash, page count and inspected render inventory.

16.2 What is independently checked

- The direct source-delta checker compares every unchanged row and coordinate, checks the weighted or quadratic growth definitions and actual frame retention, then checks every

transported operation, common-Hessian direction and full inverse map. It recollects all final quartics and verifies dimension, residual count, support, exact degree, height and coordinate participation.

- The separate reciprocal checker authenticates the complete $q + 1 = 3z + 2$ substitution, protected pool coordinates, both moduli, all integral chart/projection maps, the exact merged residual and every collected degree-ten coefficient. The independent audit rejects seven corruption classes. The component `reciprocal/` explicitly requires the bound sibling `signed/`; it is not a standalone subtree with missing inherited files.
- The separate local-majorant checker follows the literal monic gates without importing the source producer, proving its finite exponent ledger and isolating the exact local row used in the unbounded modulus proof.
- The network checks reconstruct all divergence and equality squares, their integral half-sum and every final collected coefficient. They check all monic constructor and source pullbacks, graph occurrence coverage, logical aliases, unit anchoring and the symbolic universal lift including its hub term. Separate normalization checks independently verify every new multiplicative constant and chained scale.
- The scalar checks import no producer, verify every free-ring gate identity and complete leading form, bind all four new ports together, and independently reconstruct all language and coefficient bytes. The deep native checks include carry envelopes, complete alignment cases, origin/padding masks and the actual new controller premises. All508 single-operator corruptions of the 254 arithmetic gates are rejected. Normal and optimized-Python checks remain active with assertions disabled; byte-identical replay checks reconstruction.

These are finite proofs of exact arrays, identities and explicit numerical premises. They do not prove an unbounded Pell theorem by sampling, establish the truth of consistency, or replace the syntactic calibration. The ordinary mathematical reviews are AI-assisted independent reviews, not external human refereeing or a new formal-kernel certificate.

16.3 The retained logical boundary

The direct C222/ZF and ZF/ZFC endpoints are stated in Section 2. The positive branch separately uses the retained machine/native/refutation and exact Pell reconstruction theorems, with the full new native geometry and controller proof in the appendix. All selected source texts, immutable pins and notices are included in the combined release; relocated historical paths are resolved in closure maps. No full transitive Lean or Rocq build is asserted. These boundaries are unchanged by successful finite replay.

17 Further questions with explicit proof obligations

1. Can another growth polynomial $K(t) \geq 2^{20}t$ reduce the remaining height cost? A lower-degree index can save a coordinate, but the emitted quadratic rows, actual frame atom and cofinal choice order must all be rechecked. Reducing a coefficient below the stated budget without a sharper label bound would break product separation.
2. Can sharper literal constructor bounds reduce 2^{20} or the local majorant without reinstating a derived-label range? The relevant objects are the actual schema trees and local residual, not a loose count of constructor names. The boundary $F = L^t$ is already allowed.
3. Can the 214-variable chart be brought below its current height without losing its variable advantage? Its large coefficient is not an arithmetic necessity theorem. Exact reversible row

operations or integral radical directions are admissible, but nonunit slopes require additional divisibility and cannot simply be divided out.

4. Can joint choice of coefficient factors, product constants and chained scales improve the 493-variable network? The heuristic graph cost is not the final support. Every proposal needs the monic triangular source proof and a full coefficient-height-one collection test, including all equality and unit terms.

5. Can the newer arithmetic sources improve the height-one catalog? A source transfer must be implemented and measured; the current catalog retains the old source families. Neither a lower direct dimension nor fewer residuals guarantees fewer aliases after normalization.

6. Can Start-only geometry be shortened further while preserving the complete native interface? Clean clauses must precede alignment; a flag must not be mistaken for occupation; all carry resets, mask populations and cofinal controller conditions must be proved at the new tuple.

7. Can the positive circuits reduce degree or gates as well as coefficient storage? The whole-tuple transfer changes neither. Replacing bV by b^2 in a free chart is not licensed by its current transport bound, and early cancellation through an unproved cubic scale remains invalid. The separate eleven-integer fixed-frame bounds are irrelevant to a new positive-chart architecture unless a bridge is proved.

These are concrete open optimization directions, not claimed improvements or universal lower bounds. Negative searches recorded in the source packages apply only to their stated finite families.

References

[1] ProveIt, *Pooling exact power certificates for Diophantine consistency*, source 158, 25 October 2026. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/158-pooled-powers`.

[2] ProveIt, *Removing factorial authenticity from Diophantine consistency*, source 151, 25 October 2026. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/151-factorial-free-binomial-crt`.

[3] ProveIt, *Linear CRT radii and semantic radix recovery*, source 157, 25 October 2026. Pinned alongside the preceding source; the copied `MAIN_PROOF.md` supplies the sign bootstrap and every-prime CRT completeness.

[4] ProveIt, *Quadratic projection*, source 159, 25 October 2026. Exact common-Hessian and affine projection algorithms, with an independent integer certificate checker.

[5] ProveIt, *Reusing a semantic Pell anchor over unrestricted integers* and *Recovering nonnegative arithmetic from signed Pell orientation*, source 135, retained in source 158's provenance. Copied as `PROOF_315.md` and `ORIENTATION_FIRST_PROOF.md`.

[6] ProveIt, *Aligning the amplification exponent with a strict beta quotient range*, source 135, and source 158's *Cofinal strict bounds and a linear beta base.*

[7] ProveIt, *Square-shell encoding proof* and the literal `shell_source.py` constructor source; identified by source 158's `SHELL_ENCODING_PROOF.md` supplement and exact source hashes.

[8] ProveIt, Lean sources `KCode222.lean`, `KSeedDirect222.lean`, `KEndpoint.lean`, and `ConstructibleChoice.lean`. Exact origins and inspected copies are listed in `PROVENANCE.json`; no new full Lean build is claimed.

A Every residual of the principal equation

The following is the complete finite definition of the principal polynomial. Its independent coordinates are $x_0, \ldots, x_{214} \in \mathbb{Z}$, and

$$P(x_0, \ldots, x_{214}) = \sum_{i=0}^{153} R_i(x)^2.$$

Indices are zero based, exactly as in the literal JSON array. No register or additional quantified variable is implicit in this appendix. The adjacent collected coefficient file specifies the same polynomial.

$$\begin{aligned}
R_0 &= x_0 - x_0x_4 - x_4 + x_7 \\
R_1 &= -x_2 - x_4x_5 + x_6x_7 \\
R_2 &= -1 + x_1 - x_6x_7 \\
R_3 &= 1 + x_2 - x_3x_{18} + 2x_{16} \\
R_4 &= -1 + x_8 - x_8^2 - 2x_8x_9 - x_9 - x_9^2 \\
&\quad + x_{19} \\
R_5 &= -1 - 24x_8 - 12x_8^2 - 8x_8x_9 - 16x_8x_{19} - 6x_9 \\
&\quad - 8x_9x_{19} - 14x_{19} - 4x_{19}^2 + x_{20} \\
R_6 &= -2 - x_8 - x_8^2 - 4x_8x_{20} - 4x_{20} - 4x_{20}^2 \\
&\quad + x_{21} \\
R_7 &= 1 + 2x_{21} - 4x_{21}^2 + x_{22} \\
R_8 &= -2 - x_9^2 - 2x_9x_{10} - x_{10} - x_{10}^2 + x_{23} \\
R_9 &= -1 + x_8 - x_8^2 - 2x_8x_{10} - x_9 - x_{10}^2 \\
&\quad + x_{24} \\
R_{10} &= 15 + 6x_8 - 32x_8^2 + 4x_8x_9 - 18x_8x_{10} - 16x_8x_{19} \\
&\quad - 4x_8x_{23} - 32x_8x_{24} + 25x_9 + 4x_9^2 - 8x_9x_{10} + 8x_9x_{19} \\
&\quad + 8x_9x_{23} - 3x_{10} + 9x_{10}^2 - 16x_{10}x_{19} - 8x_{10}x_{23} - 16x_{10}x_{24} \\
&\quad + 12x_{19} - 8x_{19}x_{24} - x_{20} + 4x_{23}^2 - 21x_{24} - 4x_{24}^2 \\
&\quad + x_{25} - 2x_{43} \\
R_{11} &= 14x_8 - 2x_8^2 - 8x_8x_9 + 8x_8x_{10} - 8x_8x_{23} + 24x_9 \\
&\quad + x_9^2 + 34x_9x_{10} - 16x_9x_{23} - 21x_{10} - 35x_{10}^2 + 16x_{10}x_{23} \\
&\quad + 24x_{10}x_{25} + 15x_{23} - 8x_{23}^2 - 4x_{25}^2 + x_{26} \\
R_{12} &= 2 - 8x_8 - 4x_8^2 - 8x_8x_9 - 8x_8x_{10} + 8x_8x_{23} \\
&\quad + 8x_8x_{24} - 6x_9 + 8x_9x_{23} + 8x_9x_{24} - x_{10} + 8x_{10}x_{23} \\
&\quad + 8x_{10}x_{24} - x_{19} + 4x_{23} - 4x_{23}^2 - 8x_{23}x_{24} + 6x_{24} \\
&\quad - 4x_{24}^2 + x_{27} \\
R_{13} &= 1 - 4x_8 + 2x_9 - 4x_9^2 - 8x_9x_{10} - 8x_9x_{27} \\
&\quad + 4x_{10} - 4x_{10}^2 - 8x_{10}x_{27} - 2x_{19} + 4x_{27} - 4x_{27}^2 \\
&\quad + x_{28} \\
R_{14} &= 39 - 30x_8 - 8x_8^2 - 10x_8x_9 - 20x_8x_{10} - 4x_8x_{23} \\
&\quad - 16x_8x_{24} - 22x_9 - 19x_9^2 - 22x_9x_{10} - 24x_9x_{23} + 4x_9x_{24} \\
&\quad - 27x_{10} - 19x_{10}^2 - 24x_{10}x_{23} - 16x_{10}x_{24} + 3x_{19} - 33x_{23} \\
&\quad - 4x_{23}^2 - 20x_{24} - 4x_{24}^2 + x_{29} \\
R_{15} &= 12 + 26x_8 + 26x_8^2 + 4x_8x_9 + 28x_8x_{10} + 8x_8x_{23} \\
&\quad + 32x_8x_{24} - 18x_9 - 16x_9x_{23} - 8x_9x_{24} + 36x_{10} - 18x_{10}^2 \\
&\quad + 16x_{10}x_{23} + 32x_{10}x_{24} - 24x_{10}x_{25} - 32x_{10}x_{29} - 6x_{19} - 8x_{23}^2 \\
&\quad + 30x_{24} + 8x_{24}^2 + 4x_{25}^2 - x_{26} - 2x_{29} - 4x_{29}^2 \\
&\quad + x_{30} + 4x_{43} \\
R_{16} &= -3x_8 - 47x_9 - 30x_9^2 + 12x_9x_{10} + 24x_9x_{19} + 24x_9x_{23} \\
&\quad + 5x_{10} + 6x_{10}^2 + 16x_{19} - 4x_{19}^2 - 8x_{19}x_{23} + 8x_{23} \\
&\quad - 4x_{23}^2 - x_{24} + x_{31} \\
R_{17} &= -3 - 6x_8 + 2x_9 - 2x_{10} - 2x_{24} - 4x_{31}
\end{aligned}$$

$$
\begin{aligned}
&\quad - 4x_{31}^2 + x_{32} \\
R_{18} &= 21x_8 - 8x_8^2 - 2x_8x_9 - 32x_8x_{19} + 36x_8x_{20} + 12x_9 \\
&\quad - 6x_9x_{10} - 24x_9x_{19} + 32x_9x_{20} - 15x_{10} - 3x_{10}^2 + 3x_{19} \\
&\quad - 12x_{19}^2 + 32x_{19}x_{20} - 9x_{20} - 28x_{20}^2 + 6x_{21} - 3x_{23} \\
&\quad - 3x_{24} + x_{33} + 12x_{36} - 12x_{36}^2 + 3x_{37} \\
R_{19} &= 1 + 2x_{33} - 4x_{33}^2 + x_{34} \\
R_{20} &= -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35} \\
R_{21} &= -6 - 8x_8 - 24x_8^2 + 16x_8x_9 - 32x_8x_{19} - 19x_9^2 \\
&\quad + 10x_9x_{10} + 32x_9x_{19} + x_{10}^2 - 7x_{19} - 16x_{19}^2 - 5x_{23} \\
&\quad + x_{24} + x_{36} + 2x_{71} \\
R_{22} &= -4 - 3x_8 - 7x_9 - 4x_{10} - 2x_{23} - x_{24} \\
&\quad + 4x_{36} - 4x_{36}^2 + x_{37} \\
R_{23} &= -6x_8 + 2x_9 + 2x_{10} - 4x_{10}^2 - 8x_{10}x_{37} - 2x_{24} \\
&\quad + 4x_{37} - 4x_{37}^2 + x_{38} \\
R_{24} &= -x_{10} - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39} \\
R_{25} &= -2 + 2x_8 - x_8^2 - 4x_8x_{39} - 16x_9 + 27x_9^2 \\
&\quad + 6x_9x_{10} + 24x_9x_{23} - 33x_{10} - 9x_{10}^2 + 24x_{10}x_{24} - 9x_{23} \\
&\quad + 8x_{23}^2 - 8x_{23}x_{24} + 22x_{24} - 4x_{24}^2 + 2x_{39} - 4x_{39}^2 \\
&\quad + x_{40} - 3x_{43} \\
R_{26} &= 4x_8 - 2x_8^2 - 8x_8x_{39} - 16x_{10}^2 - 16x_{10}x_{40} + 4x_{39} \\
&\quad - 8x_{39}^2 - 4x_{40}^2 + x_{41} \\
R_{27} &= -16x_8^2 + 24x_8x_{10} - 16x_8x_{19} + 16x_8x_{27} + x_{10} - 9x_{10}^2 \\
&\quad + 12x_{10}x_{19} - 12x_{10}x_{27} - 4x_{19}^2 + 8x_{19}x_{27} - 4x_{27}^2 + x_{42} \\
R_{28} &= -1 + 6x_9 - 4x_9^2 + 8x_9x_{10} - 8x_9x_{23} - 6x_{10} \\
&\quad - 4x_{10}^2 + 8x_{10}x_{23} + 6x_{23} - 4x_{23}^2 + x_{43} \\
R_{29} &= -1 - x_{10} - x_{10}^2 - 4x_{10}x_{43} - 4x_{43}^2 + x_{44} \\
R_{30} &= 1 + x_9 - x_9^2 - 4x_9x_{44} - 4x_{44}^2 + x_{45} \\
R_{31} &= -1 - 6x_{45} - 4x_{45}^2 + x_{46} \\
R_{32} &= -1 - x_{10} - x_{10}^2 - 4x_{10}x_{46} - 4x_{46}^2 + x_{47} \\
R_{33} &= 2 + x_9 - x_9^2 - 4x_9x_{47} - 4x_{47}^2 + x_{48} \\
R_{34} &= -2x_{42} - 4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49} \\
R_{35} &= -9 + 7x_8 - x_8^2 - 4x_8x_{49} + 12x_{49} - 4x_{49}^2 \\
&\quad + x_{50} \\
R_{36} &= -1 - 6x_{50} - 4x_{50}^2 + x_{51} \\
R_{37} &= -x_{10} - x_{10}^2 - 4x_{10}x_{51} - 4x_{51}^2 + x_{52} \\
R_{38} &= -x_8 - x_8^2 - 4x_8x_{52} - 4x_{52} - 4x_{52}^2 + x_{53} \\
R_{39} &= -10 - x_9 + x_{10} - x_{23} - 14x_{53} - 4x_{53}^2 \\
&\quad + x_{54} \\
R_{40} &= 1 - 2x_9 + 2x_{10} - 2x_{23} + 4x_{54} - 4x_{54}^2 \\
&\quad + x_{55} \\
R_{41} &= -10 + x_9 - x_9^2 - 4x_9x_{55} - 14x_{48} - 4x_{48}^2 \\
&\quad - 4x_{55}^2 + x_{56} \\
R_{42} &= 21 + 28x_{48} + 8x_{48}^2 - 2x_{56} - 4x_{56}^2 + x_{57} \\
R_{43} &= 2 + 15x_9 - 35x_9^2 + 14x_9x_{10} + 24x_9x_{19} - 4x_{10}x_{19} \\
&\quad - 5x_{19} - 4x_{19}^2 - x_{23} + x_{58} \\
R_{44} &= 1 + 6x_9 + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 2x_{19} \\
&\quad - 4x_{57}^2 + 4x_{58} - 4x_{58}^2 + x_{59} \\
R_{45} &= -2 - 2x_{10} + 2x_{10}^2 + 8x_{10}x_{57} + 8x_{57}^2 - 2x_{59} \\
&\quad - 4x_{59}^2 + x_{60} \\
R_{46} &= -2 - 4x_{60} - 4x_{60}^2 + x_{61} \\
R_{47} &= -x_2x_{13} + x_{62} \\
R_{48} &= x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}
\end{aligned}
$$

$$
\begin{aligned}
R_{49} &= 12x_{10}x_{62} + x_{12} - x_{13} - 2x_{26}x_{62} - x_{64} \\
R_{50} &= -2x_{28}x_{62} + 2x_{41}x_{62} - 2x_{62} - x_{65} + x_{70} \\
R_{51} &= -12x_{10}x_{62} + x_{12} - x_{13} - 2x_{30}x_{62} - x_{66} \\
R_{52} &= x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67} \\
R_{53} &= x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68} \\
R_{54} &= x_{12} - x_{13} - 2x_{38}x_{62} - x_{69} \\
R_{55} &= x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70} \\
R_{56} &= -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 \\
&\quad + x_{71} \\
R_{57} &= 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72} \\
R_{58} &= -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 \\
&\quad + x_{73} - x_{74}^2 \\
R_{59} &= -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{74} \\
R_{60} &= -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77} \\
R_{61} &= -1 + 2x_2 + 6x_2x_9 - 2x_2x_{19} + x_{75} \\
R_{62} &= -x_8x_{208} - x_{12} + x_{76} + x_{208} \\
R_{63} &= -4x_8x_{62} - 2x_9x_{62} + x_9x_{208} + x_{12} - x_{13} - 2x_{19}x_{62} \\
&\quad + x_{78} \\
R_{64} &= -x_{12}x_{63} + x_{13}x_{63} + x_{79} \\
R_{65} &= -x_{64}x_{79} + x_{80} \\
R_{66} &= -x_{65}x_{80} + x_{81} \\
R_{67} &= -x_{66}x_{81} + x_{82} \\
R_{68} &= -x_{67}x_{82} + x_{83} \\
R_{69} &= -x_{68}x_{83} + x_{84} \\
R_{70} &= -x_{69}x_{84} + x_{85} \\
R_{71} &= -x_{70}x_{85} + x_{86} \\
R_{72} &= -x_{77}x_{86} + x_{87} \\
R_{73} &= -x_{78}x_{87} + x_{88} \\
R_{74} &= -x_{88}^2 + x_{89} \\
R_{75} &= -x_4 - 2x_4x_{16} + 2x_{16} + x_{90} \\
R_{76} &= -x_{14}x_{90} + x_{91} \\
R_{77} &= -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 \\
&\quad - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{94} - x_{73}x_{89} - x_{91}^2 \\
&\quad - x_{92}^2 \\
R_{78} &= -x_4x_{15} + x_{15} - x_{15}x_{90} + x_{92} \\
R_{79} &= -x_{95}x_{98} + x_{98} + x_{100} \\
R_{80} &= -x_3x_{121} + x_{96}x_{200} - x_{99} - x_{100}x_{121} \\
R_{81} &= x_{101}x_{200} - x_{102}x_{124} - x_{103} \\
R_{82} &= x_{104}x_{200} - x_{105}x_{124} - x_{106} \\
R_{83} &= x_{107}x_{200} - x_{108}x_{124} - x_{109} \\
R_{84} &= x_{110}x_{200} - x_{111}x_{124} - x_{112} \\
R_{85} &= x_{113}x_{201} - x_{115}x_{124} - x_{116} \\
R_{86} &= x_{117}x_{201} - x_{118}x_{124} - x_{119} \\
R_{87} &= -x_{17}x_{121} + x_{124} \\
R_{88} &= x_{120} - x_{122}x_{124} - x_{123} \\
R_{89} &= -x_{124}x_{126} + x_{125} - x_{127} \\
R_{90} &= -x_{124}x_{129} + x_{128}x_{200} - x_{130} \\
R_{91} &= x_{131} - 2x_{131}x_{133} + x_{134} \\
R_{92} &= -x_{131}x_{134} - x_{134} + x_{135} \\
R_{93} &= 2x_{132} + x_{132}^2 + x_{134}^2 - x_{135}^2 \\
R_{94} &= -x_{132}x_{137} - x_{137} + x_{138} \\
R_{95} &= -1 + x_{136}^2 + 2x_{136}x_{138} + x_{137}^2 \\
R_{96} &= 1 - 2x_0 + x_0^2 - 2x_0x_{132} + x_{139} \\
R_{97} &= -x_0x_{182} + x_2 + x_{139}x_{140} - x_{181} \\
R_{98} &= 1 - 2x_3 + x_3^2 - 2x_3x_{132} + x_{141}
\end{aligned}
$$

$$
\begin{aligned}
R_{99} &= -x_3x_{173} + x_{17} + x_{141}x_{142} - x_{172} \\
R_{100} &= -1 + 2x_{132}x_{184} - x_{143} + 2x_{184} - x_{184}^2 \\
R_{101} &= -1 + x_{95} + x_{143}x_{144} - x_{172} - x_{173}x_{184} \\
R_{102} &= -x_{132}x_{146} - x_{146} + x_{147} \\
R_{103} &= -1 + x_{145}^2 + 2x_{145}x_{147} + x_{146}^2 \\
R_{104} &= 1 + x_{132} - x_{132}x_{149} - x_{149} + x_{150} \\
R_{105} &= x_{148}^2 + 2x_{148}x_{150} - 2x_{149} + x_{149}^2 \\
R_{106} &= -x_{132}x_{152} - x_{152} + x_{153} \\
R_{107} &= -1 + x_{151}^2 + 2x_{151}x_{153} + x_{152}^2 \\
R_{108} &= -x_{132}x_{155} - x_{155} + x_{156} \\
R_{109} &= -1 + x_{154}^2 + 2x_{154}x_{156} + x_{155}^2 \\
R_{110} &= -x_{132}x_{158} - x_{158} + x_{159} \\
R_{111} &= -1 + x_{157}^2 + 2x_{157}x_{159} + x_{158}^2 \\
R_{112} &= -x_{132}x_{161} - x_{161} + x_{162} \\
R_{113} &= -1 + x_{160}^2 + 2x_{160}x_{162} + x_{161}^2 \\
R_{114} &= -x_{132}x_{164} - x_{164} + x_{165} \\
R_{115} &= -1 + x_{163}^2 + 2x_{163}x_{165} + x_{164}^2 \\
R_{116} &= -x_{132}x_{167} - x_{167} + x_{168} \\
R_{117} &= -1 + x_{166}^2 + 2x_{166}x_{168} + x_{167}^2 \\
R_{118} &= -x_{132}x_{170} - x_{170} + x_{171} \\
R_{119} &= -1 + x_{169}^2 + 2x_{169}x_{171} + x_{170}^2 \\
R_{120} &= 1 - x_{132}x_{173} - x_{173} + x_{174} \\
R_{121} &= -1 + 2x_{172} + x_{172}^2 + 2x_{172}x_{174} + x_{173}^2 \\
R_{122} &= -x_{132}x_{176} - x_{176} + x_{177} \\
R_{123} &= -1 + x_{175}^2 + 2x_{175}x_{177} + x_{176}^2 \\
R_{124} &= -x_{132}x_{179} - x_{179} + x_{180} \\
R_{125} &= -1 + x_{178}^2 + 2x_{178}x_{180} + x_{179}^2 \\
R_{126} &= -x_{132}x_{182} - x_{182} + x_{183} \\
R_{127} &= -1 + x_{181}^2 + 2x_{181}x_{183} + x_{182}^2 \\
R_{128} &= -5 - 2x_0 - 4x_0^2 - 2x_0x_2 - x_1^2 - 2x_2^2 \\
&\quad - x_3^2 - x_4^2 + 2x_8 - x_8^2 - x_9^2 - x_{10}^2 \\
&\quad - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - 4x_{16}^2 \\
&\quad - x_{17}^2 - x_{93}^2 + 2x_{95} - 2x_{95}^2 - x_{96}^2 - x_{97}^2 \\
&\quad - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 - x_{114}^2 \\
&\quad - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 + x_{131} \\
&\quad - x_{184}^2 \\
R_{129} &= 2x_3 - 4x_3x_8 - 4x_3x_9 - 4x_3x_{10} - 4x_3x_{11} - 4x_3x_{12} \\
&\quad - 4x_3x_{13} - 4x_3x_{14} - 4x_3x_{15} - 8x_3x_{16} + x_{184} \\
R_{130} &= -32 + x_0 - 4x_{185}^2 \\
R_{131} &= -x_{132}x_{187} - x_{187} + x_{196} \\
R_{132} &= -1 + x_{186}^2 + x_{187}^2 - x_{196}^2 \\
R_{133} &= -x_{189} - x_{189}x_{190} + x_{197} \\
R_{134} &= -1 + x_{188}^2 + x_{189}^2 - x_{197}^2 \\
R_{135} &= -x_{179}^2 + x_{198} \\
R_{136} &= x_{187} - 2x_{192}x_{198} - x_{198} \\
R_{137} &= -4x_{187} - 8x_{187}x_{191} + x_{190} \\
R_{138} &= -x_{132} - x_{186}x_{193} + x_{190} \\
R_{139} &= -x_{178} - x_{180} - x_{186}x_{194} + x_{188} \\
R_{140} &= -2x_{16} - 4x_{179}x_{195} + x_{189} \\
R_{141} &= 26 + 2x_0^2 + 2x_0x_2 - 2x_0x_{137} - 2x_0x_{146} + x_1^2 \\
&\quad - 2x_1x_{173} + x_2^2 - 2x_2x_{146} + x_4^2 - 2x_4x_{164} + x_8^2 \\
&\quad - 2x_8x_{149} + x_9^2 - 2x_9x_{152} + x_{10}^2 - 2x_{10}x_{155} + x_{11}^2 \\
&\quad - 2x_{11}x_{158} + x_{12}^2 - 2x_{12}x_{161} + x_{13}^2 - 2x_{13}x_{167} + x_{14}^2
\end{aligned}
$$

$$
\begin{aligned}
&- 2x_{14}x_{170} + x_{15}^2 - 2x_{15}x_{176} + 4x_{16}^2 - 4x_{16}x_{179} + x_{93}^2 \\
&- 2x_{93}x_{182} - x_{132}x_{199} - 2x_{136} - 2x_{138} + x_{138}^2 - 2x_{145} \\
&- 2x_{147} + x_{147}^2 - 2x_{148} + 2x_{149} - 2x_{150} + x_{150}^2 \\
&- 2x_{151} - 2x_{153} + x_{153}^2 - 2x_{154} - 2x_{156} + x_{156}^2 \\
&- 2x_{157} - 2x_{159} + x_{159}^2 - 2x_{160} - 2x_{162} + x_{162}^2 \\
&- 2x_{163} - 2x_{165} + x_{165}^2 - 2x_{166} - 2x_{168} + x_{168}^2 \\
&- 2x_{169} - 2x_{171} + x_{171}^2 - 2x_{172} + x_{174}^2 - 2x_{175} \\
&- 2x_{177} + x_{177}^2 - 2x_{178} - 2x_{180} + x_{180}^2 - 2x_{181} \\
&- 2x_{183} + x_{183}^2 \\
R_{142} &= -x_{97}x_{200} + x_{121} \\
R_{143} &= -x_{114}x_{201} + x_{121} \\
R_{144} &= -2 - 4x_{16}^2 - x_{137}^2 - x_{146}^2 + 2x_{149} - x_{149}^2 \\
&- x_{152}^2 - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 \\
&- x_{170}^2 - x_{173}^2 - x_{176}^2 + x_{179}^2 - x_{182}^2 - x_{202}^2 \\
&- x_{203}^2 - x_{204} - x_{204}^2 \\
R_{145} &= -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 \\
&- 4x_{116}^2 - 4x_{119}^2 + x_{121}^2 - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 \\
&- x_{205}^2 - x_{206}^2 - x_{207} - x_{207}^2 \\
R_{146} &= -x_2x_{12} + x_{208} \\
R_{147} &= x_{95}x_{210} - 2x_{96}x_{146} + 2x_{101} - 2x_{101}x_{149} - 2x_{104}x_{152} - 2x_{107}x_{155} \\
&- 2x_{110}x_{158} - 2x_{113}x_{161} - 2x_{117}x_{167} - 2x_{120}x_{170} - 2x_{125}x_{176} - 2x_{128}x_{179} \\
&- 2x_{132}x_{210} + 2x_{145}x_{146} + 2x_{146}x_{147} - 2x_{148} + 2x_{148}x_{149} + 2x_{149}x_{150} \\
&- 2x_{150} + 2x_{151}x_{152} + 2x_{152}x_{153} + 2x_{154}x_{155} + 2x_{155}x_{156} + 2x_{157}x_{158} \\
&+ 2x_{158}x_{159} + 2x_{160}x_{161} + 2x_{161}x_{162} + 2x_{166}x_{167} + 2x_{167}x_{168} + 2x_{169}x_{170} \\
&+ 2x_{170}x_{171} + 2x_{175}x_{176} + 2x_{176}x_{177} + 2x_{178}x_{179} + 2x_{179}x_{180} + x_{209} \\
&- 2x_{210} \\
R_{148} &= -1 - x_{95}x_{209} + x_{96}^2 - 2x_{96}x_{145} + x_{101}^2 - 2x_{101}x_{148} \\
&+ x_{104}^2 - 2x_{104}x_{151} + x_{107}^2 - 2x_{107}x_{154} + x_{110}^2 - 2x_{110}x_{157} \\
&+ x_{113}^2 - 2x_{113}x_{160} + x_{117}^2 - 2x_{117}x_{166} + x_{120}^2 - 2x_{120}x_{169} \\
&+ x_{125}^2 - 2x_{125}x_{175} + x_{128}^2 - 2x_{128}x_{178} + x_{145}^2 - x_{146}^2 \\
&+ x_{148}^2 + 2x_{149} - x_{149}^2 + x_{151}^2 - x_{152}^2 + x_{154}^2 \\
&- x_{155}^2 + x_{157}^2 - x_{158}^2 + x_{160}^2 - x_{161}^2 + x_{166}^2 \\
&- x_{167}^2 + x_{169}^2 - x_{170}^2 + x_{175}^2 - x_{176}^2 + x_{178}^2 \\
&- x_{179}^2 + x_{210} \\
R_{149} &= x_{95}x_{212} - 2x_{97}x_{137} - 2x_{114}x_{164} - 2x_{121}x_{173} - 2x_{132}x_{212} + 2x_{136}x_{137} \\
&+ 2x_{137}x_{138} + 2x_{163}x_{164} + 2x_{164}x_{165} + 2x_{172}x_{173} + 2x_{173} + 2x_{173}x_{174} \\
&+ x_{211} - 3x_{212} \\
R_{150} &= -x_{95}x_{211} + x_{97}^2 - 2x_{97}x_{136} + x_{114}^2 - 2x_{114}x_{163} + x_{121}^2 \\
&- 2x_{121}x_{172} + x_{136}^2 - x_{137}^2 + x_{163}^2 - x_{164}^2 + x_{172}^2 \\
&- x_{173}^2 + x_{211} + x_{212} \\
R_{151} &= -1 - x_2 - 2x_2x_{213} + x_4 \\
R_{152} &= -x_0^2 + x_{214} \\
R_{153} &= x_{93} - 32x_{214}^2
\end{aligned}
$$

B The complete retained Start only native proof

This appendix retains the complete incoming ordinary native proof, dated 25 October 2026. Its fifteen-positive P66 equation is an inherited bridge, not an additional new endpoint. Historical relative paths refer to the retained native source subtree. The Round22 ten-positive transfer is proved in Section 14.

This note is an ordinary mathematical construction, composed with the retained K215 / native / positive-P66 / syntactic-refutation calibration. It changes all four fixed integer coefficients coherently. It does not assert consistency, global minimality, an end-to-end proof-assistant certificate, or worldwide novelty. The idea of trying a Start-only marker is explicitly suggested by the active repository SIZE_REDUCTION.md; this note completes a particular shorter layout, its complete proof and its finite controller certificate.

B.1 Exact source and result

The immutable comparison commit is `fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb` of `VladimirReshetnikov/ProveIt`. The immediate whole ordinary source is `Computability/DiophantineConsistency/research/zero-weight-flag-independent/PROOF.md`. The full 943-tile language, 430 actions, 436-character numeral and decoder are exactly those printed in `Computability/DiophantineConsistency/equation.tex` at that commit. Local copies are retained in `inherited/`.

The same fifteen-variable P66 free-port polynomial now has a coherent tuple with cell length 4127 instead of 6007 and binary cell size

```
d = 47 * 4127 = 193969,    B = 2^193969,    beta = B-1.
```

Its four minimal unsigned big-endian coefficient payloads occupy 97046 bytes, versus 141226, a reduction of 44180 bytes (about31.28%). There are 66 live integer-leaf arithmetic operations and exact degree157, as before. The new DC kernel has1957 nonzero rho-adic coefficients, versus2899: it deletes942 unit-kernel terms. These constant-recipe counts are distinct from the66-gate evaluation cost. No unknown or equation is hidden in the coefficient recipe.

Every assertion below uses the new tuple. Old6007 K/controller residues are never reused.

B.2 The whole language, support and coefficients

Put rho=2^47, M=2^44 and mu=M-2. Retain the old tile list `t_j=(w_j,n_j,p_j,s_j)`, j=0,...,942. Colors interpreted by the native clauses are the positive complements of these old labels: eight-bit east/west and nine-bit north/south. In particular the forced Start tile is `(2,435,3,1)`, so its complemented south label is510. Its bits1 through8 are all one. All genuine complemented colors are positive; empty cells have zero colors.

The physical positions in one rho-cell are exactly

```
E = {0,...,942} union {943,...,951} union {973,...,981}
    union {992,...,999} union {1002,...,1009} union {990,991,1031}.
```

There are980 positions. The low dummy990 can carry digits0,...,3; all other physical digits are Boolean. Position1031 is a separate zero-weight physical flag. The flag has no clause weight. Define every coefficient c_e, for 0<=e<=3296, by

```
c_j = 1 + 2^10(1+w_j+2^8 n_j+2^17 p_j+2^25 s_j), 0<=j<=942;
c_(1002+j)=2^(10+j), c_(992+j)=2^(27+j), 0<=j<8;
c_(973+j)=2^(18+j), c_(943+j)=2^(35+j), 0<=j<9;
all remaining coefficients are zero.
```

Thus c_990=c_991=c_1031=0. These3297 coefficients are pointwise identical to the old zero-weight6007 coefficient vector: changing the physical flag does not change that vector. There are977 nonzero coefficients, each<M, and the34 label coefficients sum to M-1024. The checker independently decodes every action and tile from the436-character numeral before comparing all3297 weights with the frozen parent vector.

Let G_m(x)=sum_(j=0)^(m-1)x^j and define

```
Psi(x,t)=x^992(x^10+t^17)G_8(tx)
         +t^8 x^943(x^30+t^17)G_9(tx)
         +t^34(2-t)x^1031,
Phi(x)=sum_(j=0)^942 c_j x^j + 2^10 Psi(x,2),
Lambda=rho^2042+rho^3074.
```

At t=1, Psi specifies35 physical label/flag positions. At t=2, its flag term vanishes and it gives the same weighted Phi as before. Physical support must not be inferred from nonzero Phi coefficients.

B.3 Four fixed integers, completely specified

Take L=4127 and B=rho^L. Define

```
DC = rho^20 + rho^30 + rho^1031 + Lambda*Phi(1/rho),
K = DC + B*rho^10,
MC = beta + rho - (rho+3)rho^990 - G_943(rho) - Psi(rho,1),
MF = B+3+(M-2)Lambda + rho^973 G_9(rho)
     +rho^1012 G_8(rho)+rho^1031
     +2^29 rho^4124-rho^4106.
```

Every reflected exponent is nonnegative: the first reflected coefficient is at2042-1009=1033. Hence the apparently rational expression in DC is an integer specified by a finite sum. In the P66 graph use the four ports `beta,K0=K,MC,MF`. This MF is the source mask, already shifted once by beta. Write MF_native=MF-beta only in the proof.

The core mask consists of two43-bit clause masks,17 label-copy tests, one flag-copy test and the origin test, for105 bits. The zero padding is 18 complete rho-fields4106,...,4123 and29 bits at4124, for875 more bits. The positive padding identity is

```
2^29 rho^4124-rho^4106
  = (rho-1)sum_(j=4106)^4123 rho^j + (2^29-1)rho^4124.
```

The full finite integer computation verifies

```
0<DC<B,    0<MC<beta,    0<MF_native<beta,
pc(MF_native)=980,    pc(MC)=d-980,
MC=2 mod4,    MF_native=4 mod8.
```

No carries or borrows in these explicit mask calculations are left to an informal picture. They are checked on the complete integers. These are the same numerical interface conditions used by the inherited outer arithmetic.

B.4 The initial clause lemma and bounded cyclic carries

The zero-weight clause lemma of the immediate source is unchanged. For a selector population n, the complemented-label word W and selected tile words u_j, write

```
Q = 2^34 n + W - sum_selected u_j.
```

The clause raw value is n+2^10 Q. With incoming carry in0,...,29 and the mask mu, the source's integer inequalities force n=0 or1. For n=0 all labels are zero; for n=1 exactly one selected tile determines all34 complemented labels. A flag is not recovered or used by this lemma.

The new DC has three bare unit terms, at20,30,1031, and two reflected coefficient blocks. The block separation is1032, larger than max(E)=1031. Therefore a given output has at most one predecessor for each reflected coefficient, even though two full blocks occur in DC. The shifted low-dummy extra0/2 is treated separately. It meets at most one reflected block, since their separation is again absent from E. Its unit hit2021 precedes every reflected extra-dummy hit, whose first possible field is2023. Its bare and spatial extra hits are1010,1020,1000, all distinct.

One may use the inherited hand bound on the full coefficient sum and obtain the same29 carry bound. The finite certificate also supplies an independent stronger direct check: form the full nonnegative convolution DC*1_E, add twice the translated DC at dummy990, and take the exact maximum Hmax. This is a finite array containing every permitted non-temporal raw value. The complete arithmetic check is

```
Hmax + 3 + 3rho/4 + 29 < 30rho.
```

Here3 bounds the spatial source digit; 3rho/4 bounds the temporal digit in any fractional binary phase. At phase46 the only extra odd spill is from low dummy990 into991, already in E. This bound is independent of alignment, the flag value, or a solution to the clause tests.

The monotone cyclic carry map preserves the finite box0,...,29. Iterating from zero stabilizes and telescoping identifies its digit word with the transport residue. The strict representative supplied by the inherited outer inequalities rules out the zero/all-ones ambiguity, exactly as in the source. Since the clause centers differ by1032, they cannot both have a temporal physical predecessor in any cell or fractional phase. At least one clean clause applies. Thus every cell has zero or one selector and its correct complemented labels before any alignment assertion.

B.5 A nine-target alignment lemma

The new singleton rho^1031 copies only selector0 (Start) to field1031. It does not copy all943 selectors. Below1033 there is no reflected term. There are at most three Boolean bare contributions and the bounded dummy contribution; the safe bound9 suffices. Hence

```
9 + 3rho/4 + 29 < rho
```

makes every outgoing carry in fields0,...,1032 zero. This includes every incoming carry at targets974,...,981 and1031.

In the forced Start cell, the eight targets974,...,981 each receive one from the corresponding south label. Target1031 also receives one from the Start selector. Each of their low-bit tests requires an odd temporal contribution. Fractional phases1,...,45 have only even temporal digits. Phase46 has at most one odd low-dummy spill, and cannot meet nine targets. Thus the fractional phase is zero.

Let r be the remaining rho-digit offset moduloL, and put u=974-r moduloL. The eight temporal preimages u,...,u+7 must be physical. They cannot wrap across the long unsupported top interval. Within each predecessor cell at most one of them may be a selector, by the already established clause lemma. The complete possible starts are therefore exactly

```
u in {942,943,944,973,974,990,991,992,1002}.
```

This is also immediate by inspecting the five physical intervals: at most one selector permits only942,...,944 at the end of0,...,951. The ninth temporal preimage is u+57. Among the nine displayed starts, only u=942 and u=974 have u+57 in E, namely999 and1031. The first case would require selector942 to be one and also the south low bit at943 to be one in the same predecessor cell. The clean clause already forces that cell's labels to be those of tile942. The complete decoder

gives tile942=(4,437,2,437), whose complemented south label is511-437=74. Its low bit is zero, a contradiction. Therefore u=974 and r=0 moduloL. No value of the physical flag was assumed, and no uniqueness of Start outside the forced cell was assumed. The finite checker additionally tests every r=0,...,L-1, partitioning possible wrapped preimages by their actual predecessor cells. Support and selector capacity leave offsets0,32; the decoded labels exclude32, leaving only0.

B.6 Exact copies, all resets and soundness

The full convolution identities are

```
target973+j: current943+j, 0<=j<9;
target1012+j: current992+j plus spatial1002+j, 0<=j<8;
target1031: currentselector0;
targets2042 and3074: sum_e c_e current_e.
```

The singleton cannot contaminate a clause: its would-be predecessors are 1011 and2043, both absent. Its would-be predecessor at2041 is1010, also absent. Reflection starts above all copy tests. The bare/spatial sources at all clause targets and predecessors are absent. These identities are checked as complete coefficient lists, not just at intended witnesses.

After alignment, the17 ordinary copies force every temporal/spatial color equality. The flag copy gives precisely

```
flag_(temporal predecessor of i) = StartSelector_i.
```

Since temporal rotation is a permutation, this uniquely determines all flags from Start selectors. Occupation is instead forced by color equality: empty cells have zero colors, genuine tiles have positive complemented colors, and spatial one-cell rotation propagates occupancy from the forced Start to every cell. Thus every cell contains exactly one genuine tile. The flag is never required to equal occupancy.

Let Aadj=sum_(e,e+1 in E)c_(e+1)x_e. The two complete clause predecessor values are

```
field2041: Aadj;
field3073: Aadj+c_0*x_1031.
```

The extra term comes from the unique pair1031-0=1031; difference1033 is absent. With at most one selected tile, Aadj<=2M-1024, including selector-to-label adjacency942/943 and dummy-to- label adjacency991/992. The flag has no adjacent nonzero coefficient. Since c_0<M, both predecessors plus incoming29 are<rho and reset carry. The actual occupied clause value is M+1<rho.

The maximum raw product field is3074+1031=4105. Its entire value is c_0*x_1031<M; it resets incoming29. All fields4106,...,4126 are raw zero, and aligned temporal/spatial support is absent there. This proves every875 padding bit and a terminal zero field. Each B-cell origin is reset. Origin behavior is identical to the source: DC and the spatial shift have positive exponents, so only the temporal Start selector can meet an origin.

The forced Start's complemented north76 differs from its south510, so the temporal stride is nonzero. Inverse complementation now gives exactly the retained genuine marked blank computation and syntactic ZFC refutation. Soundness uses no finite controller or prime-existence assumption.

B.7 Fresh canonical encoding and strict inequalities

From a finite syntactic refutation, the retained helical-padding theorem gives genuine one-Start rectangles with independently cofinal width and height. Complement the colors and encode all cells on the new E. Define each flag by the unique permutation rule in Section6. Supply both

dummies initially zero. Every Boolean subset of safe low990 dummy bits leaves all selectors, colors and the flag relation unchanged.

The same bounded cyclic carry construction is valid before choosing a strict representative. The exact aligned resets proved above apply to every such subset. Occupied clauses make F positive, and the terminal zero field gives

```
0<F<q/rho<q/7,
0<C<q/rho^(L-1032),
2C+F<q*(1/7+2/rho^(L-1032))<q/3.
```

Thus alpha=q-F-C is positive, and0<F<q-1. The flags have been encoded afresh; the old6007 word with all flags one is not reused.

B.8 Actual rank and the new controller

Set t=dN, q=2^t, J=(q-1)/beta and

```
S'=(C-1)+qF,
Tmask=(MC*J+1)+q*(MF_native*J-1).
```

The inherited outer mask theorem gives disjoint S' and Tmask belowq^2. The explicit populations and origin changes give pc(Tmask)=t+2. With exactly one source-mask shift, the actual P66 rank is

```
R=[q(q-F)-C](q^2-1)+(MC+qMF)J
  =(q^2-1-S')q^2+(S'+Tmask).
```

It follows that pc(R)=3t+2. The unchanged origin residues C=1 mod4, F=0 mod4 and J=1 mod4 give R=3 mod4.

A safe bit at cellj changes C by rho^990 B^j. The complete DC/dummy maximum exponent is3074+990=4064. Its other later-cell exponents are1000 (spatial) and990 (temporal). All remain belowL with at least three free rho-digits. Different later cells or distinct later exponents prevent coefficient collisions. Exactly as in the source, for j+1<N and j+width<N one obtains a positive ordinary increment deltaRaw<q/7 and, uniformly over intermediate subsets, F_old+deltaRaw<2q/7<q-1. Thus no modular wrap is hidden in

```
deltaR=-Gamma B^j,
Gamma=rho^990(q^2-1)[1+q(K+B^width)].
```

The new finite certificate gives the following exact controller data:

```
L=4127 prime, d=193969,
ord_47(2)=ord_47(B)=23,
ord_4127(2)=ord_4127(B)=2063,
spacing=lcm(23,2063)=47449,
lambda=474491=10*47449+1 prime > d.
```

All orders are checked by their full exponent and every distinct prime factor of that exponent. Both orders are coprime to d. Also B^2!=1 modulo each of47,4127,lambda, and the certificate gives

```
H0=1+B*K+B^2 = 9 mod47,
H0 = 3487 mod4127,
H0 = 153928 mod474491,
B^(lambda-1) = 1+135285*lambda mod lambda^2,
0 < 135285 < lambda.
```

The precise last residue is in `start_only_certificate.json`. Primality here is established by full trial division up to the square root, so no probabilistic primality or unsupported prime assertion is used. The dense Phi evaluation, sparse DC evaluation and complete integer K agree at all three primes.

For completeness choose independent large width=lambda^r_w and height=lambda^r_h, with N=width*height=lambda^n, N>d*lambda^2 and height>lambda. Put T_orbit=N/lambda and C0=d*lambda. Every power of lambda is1 modulo lambda-1, so q=B^N=B^width=B at every prime divisor ofd*lambda. The new units above make Gamma a unit modulodN. The valuation-one lift gives exact orderT_orbit ofB^(lambda-1) modulo lambda^n, while that power is1 modulod. CRT gives its orbit as1+C0*z, 0<=z<T_orbit.

For clarity, the source's subset construction is complete with these new parameters: normalize a target byB^-2 and choose epsilon in{0,1} so that subtracting epsilon*B leaves a nonzero residue modulo lambda. Its least representative a0 moduloC0 lies in1,...,C0-1 and is a unit moduloT_orbit. Write the target a0+C0*u and solve

```
a0*z0+a0*(a0-1)/2 = u modT_orbit.
```

The a0 consecutive distinct orbit coordinates sum to the target. Use the safe bit indices2+(lambda-1)j and the additional index3 exactly when epsilon=1. Their largest index is N-T_orbit-lambda+3; adding1 or width stays belowN. Applying this subset to (R0-d*width)*Gamma^-1 modulo dN gives

```
R=d*width mod dN.
```

All masks and inequalities are preserved. The source's rank growth yields R>q(q^2-1)>dN+d*width. Hence the required positive outer power and transport witnesses are constructed exactly as before with the actual new K.

B.9 P66 and the final consistency equivalence

Use precisely the fifteen independent positive integer variables

```
J,F,alpha,z,f,h,i,T,s,w,tau,eta,zeta,y,gamma.
```

All other names in the following display are polynomial abbreviations:

```
q_-=beta*J, q=q_-+1, C=q-F-alpha,
X=wq, Y=sq^3, E=XY, k=eta+zeta,
ell=EkY, c=kY+eta, a=E+Y, H=4a+3,
Delta=a^2+H, D=X+ac+gamma*H,
R=[q_-(q-F)+alpha](q^2-1)+(MC+qMF)J,
V=f^2(cT-R)-c, S=i*Delta*c^2,
N0=tau^2-ell(ell+k), N1=D^2-Delta*c^2,
Na=S^2(V^2-y^2)+y^2, Nk=k-hE-R,
Nt=(K+w)C+(q-F)-z*q_-, Ns=Delta*f^2-S^2,
P=N0*N1*Na*Nk*Nt*Ns-Delta.
```

The single Diophantine equation is P=0. The complete source's Section12 establishes both P66 witness maps for any coherent native tuple with the scalar boxes, masks, genuine native soundness and cofinal controller completeness proved in Sections3–8 above. None of its Pell constructions depends on the old flag location or on setting all flags one. Its proof uses the exact rank identities pc(R)=3t+2, R=3 mod4, its canonical bounds and the actual native transport; all have just been reconstructed for this tuple. Its signed unit analysis, main/strong/auxiliary index recovery, half-binomial projection and positive completeness maps therefore apply unchanged, with the new four coefficients together.

Consequently ordinary ZFC proves

```
(there exist fifteen positive integers with P=0)
   iff (there exists a finite syntactic ZFC refutation).
```

Negating yields unsolvability iff Con(ZFC). The ordinary proof does not decide either side. It composes the retained explicit foundational and machine calibration, rather than replacing them by the numerical box conditions alone.

The literal free-port P66 graph is unchanged. Its66 live integer gates, fifteen live positive inputs and exact factor degrees (22,18,62,7,2,46) are retained. The full leading form has the nonzero monomial coefficient8*beta^92, so the output degree remains exactly157 at the new positive beta. Different fixed tuples give different specialized polynomials; no equality with the old6007 specialization is claimed.

B.10 Verification and useful next questions

The previously retained4159 construction uses physical flag1042 and a support-only nine-target alignment argument. This4127 refinement additionally uses the actual decoded color of tile942; it saves752 further raw bytes. The two coefficient tuples and their controller certificates are distinct.

`python3 start_only.py -write` deterministically creates the complete four binary coefficients, all943 tiles, all3297 weights and the finite certificate. `python3 start_only.py` reconstructs them without writing. Python optimization does not remove guards: every check uses an explicit exception. The checker reads the frozen coefficient vector as data, but imports no inherited implementation and separately decodes the whole seed. It checks full convolution lists at every designated field, every possible digit-offset alignment, the actual carry envelope, all mask bits, every controller order/unit/lift and every coefficient byte.

The unbounded proof is Sections4–9, not the finite tests. An independent review should especially challenge the order: clean clauses first, then at-most-one-selector, then nine-target alignment, then flag interpretation and occupied-cell propagation. Treating the flag as occupancy before alignment would reinstate the unnecessary943-term kernel.

Further genuine opportunities are a shorter separated label layout, a smaller guaranteed-zero padding cost, or another physical alignment pattern with fewer marker tests. They require complete new geometric and controller proofs. This note supplies no global lower bound for cell length, coefficient storage, polynomial degree, variable count or arithmetic operations.